Visualization Fundamentals

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Slides credited to
Jessica Hullman

The big picture

task

e.g., find outliers,
diagnose model fit,
make decision

data

physical type

int, float, etc.

abstract type

nominal, ordinal, etc.

domain

metadata
semantics
conceptual model

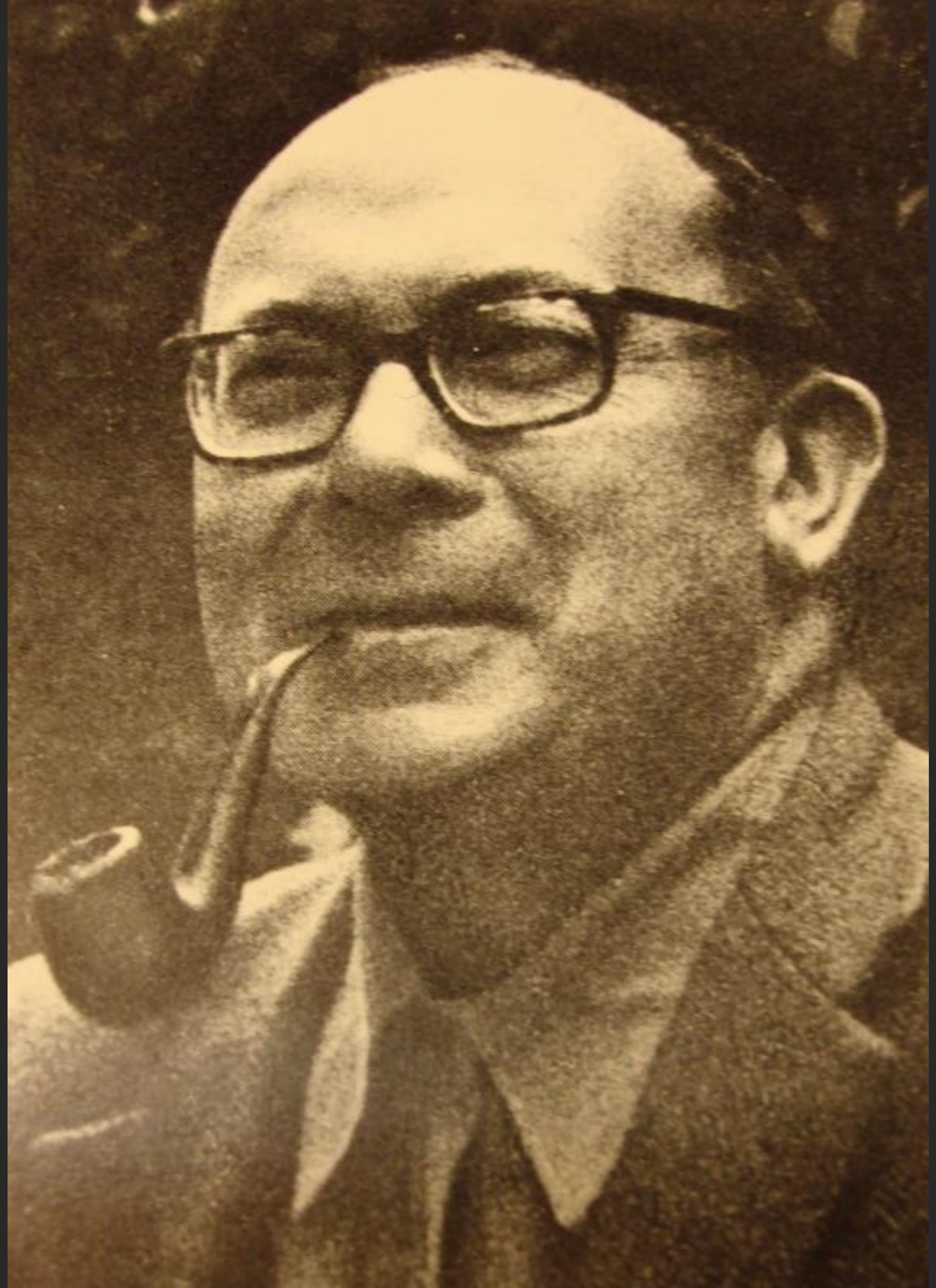
processing
algorithms

mapping

visual encoding
visual metaphor

image

visual channel
retinal variables



Visual language is a sign system

Jacques Bertin

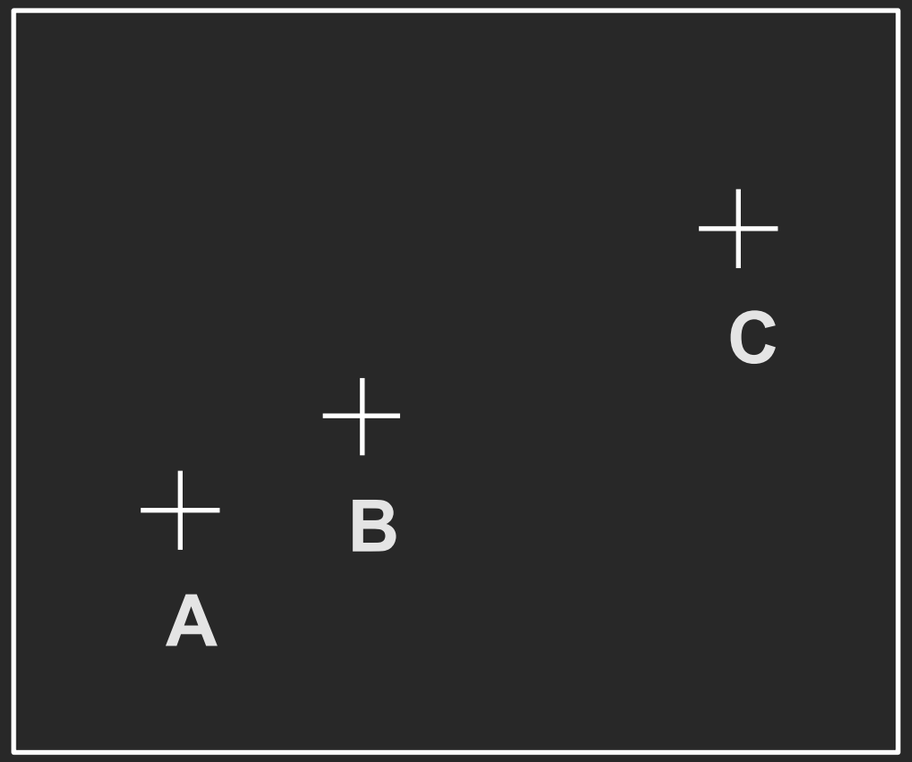
Images perceived as a set of signs

Sender encodes information in signs

Receiver decodes information from signs

Semiology of Graphics, 1967

Information in position



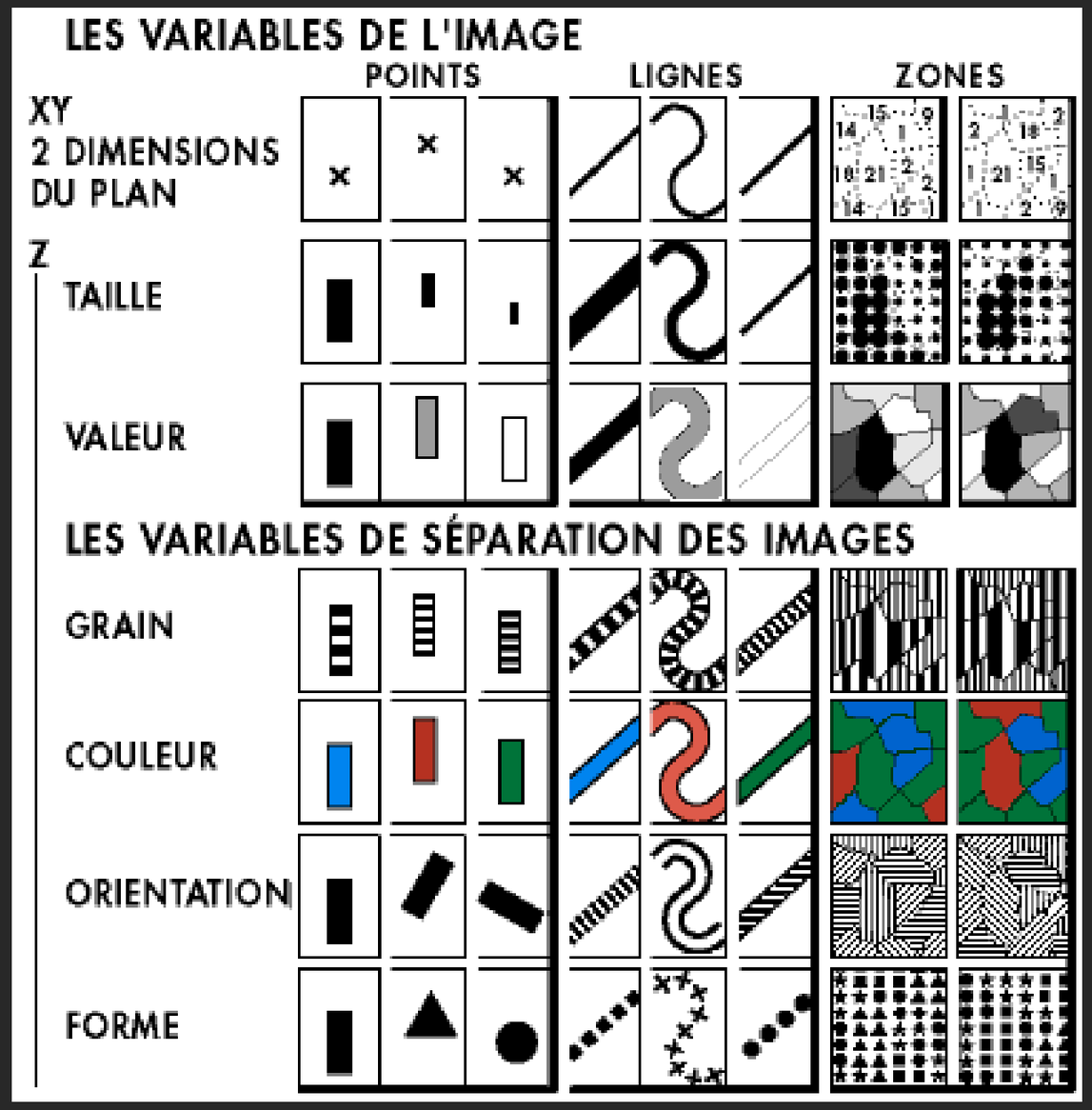
1. A, B, C are distinguishable
2. B is between A and C.
3. BC is twice as long as AB.

∴ Encode quantitative variables

"Resemblance, order and proportional are the three signfields in graphics." - Bertin

Visual variables

- Position (x 2)
- Size
- Value
- Texture
- Color
- Orientation
- Shape

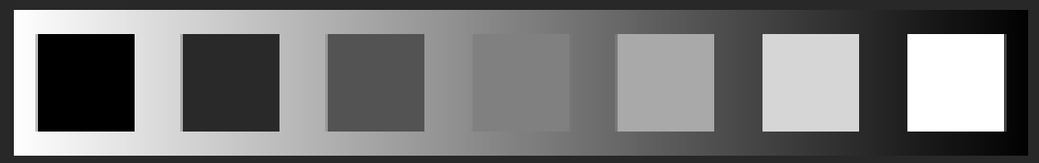


Note: Bertin does not consider 3D or time
Note: Card and Mackinlay extend the number of vars.

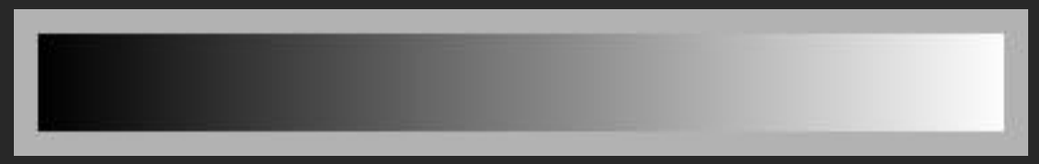
Information in color and value

Value is perceived as ordered

∴ Encode ordinal variables (O)



∴ Encode continuous variables (Q) [not as well]



Hue is normally perceived as unordered

∴ Encode nominal variables (N) using color

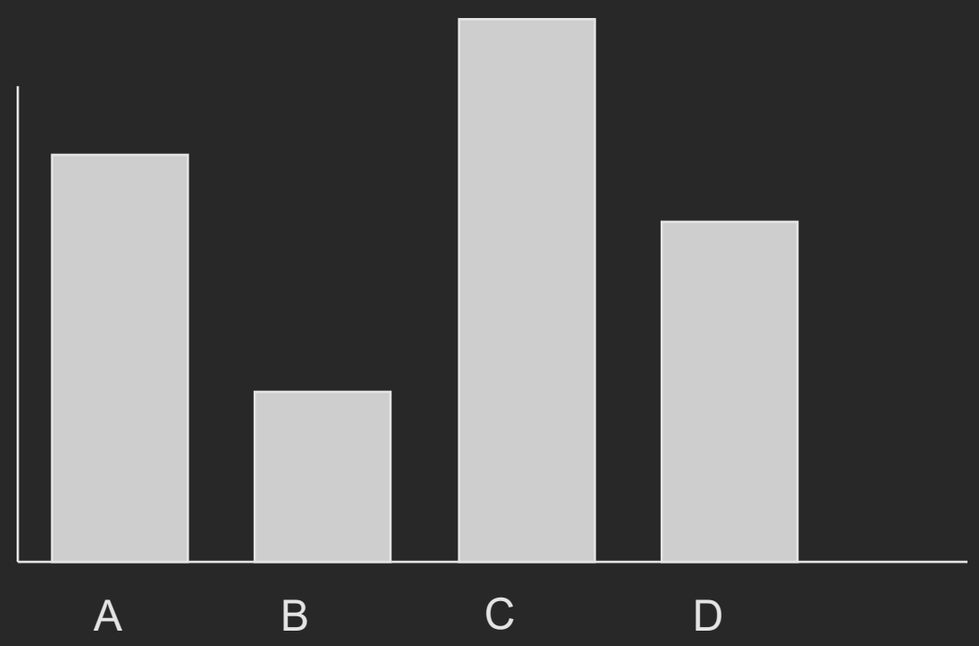
Encodings as constraints

Univariate data

variables

observations

	v1	v2	v3	
A				
B				
C				

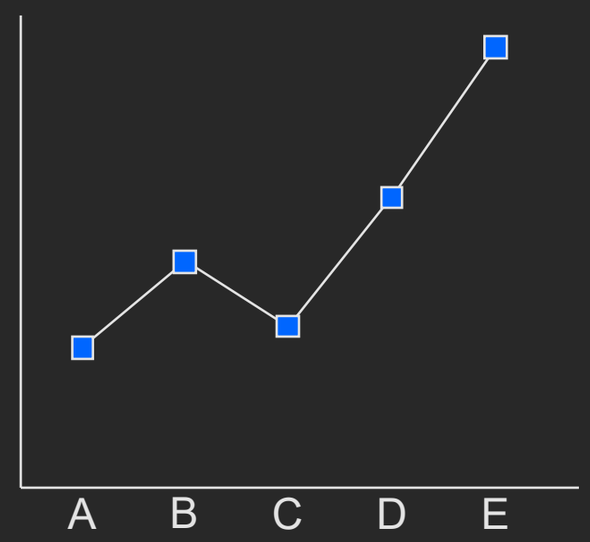


Univariate data

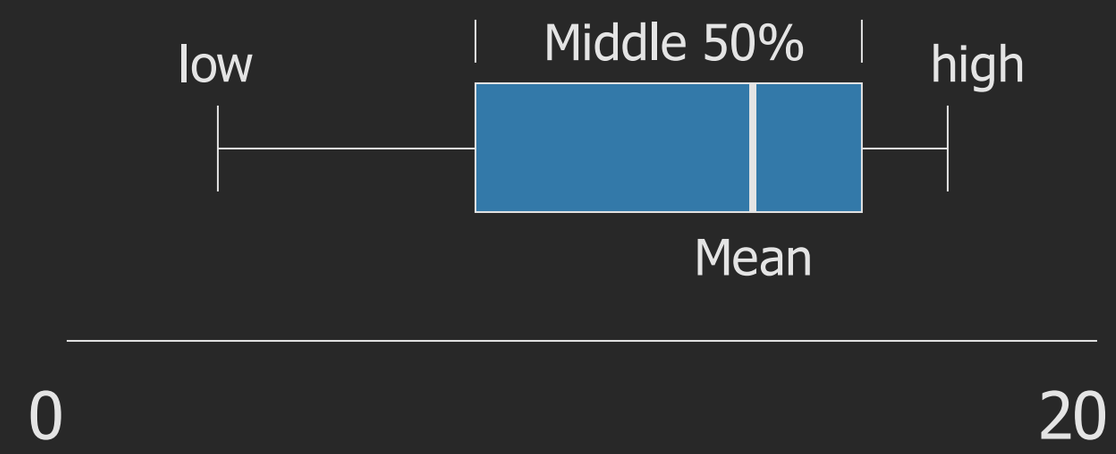
variables

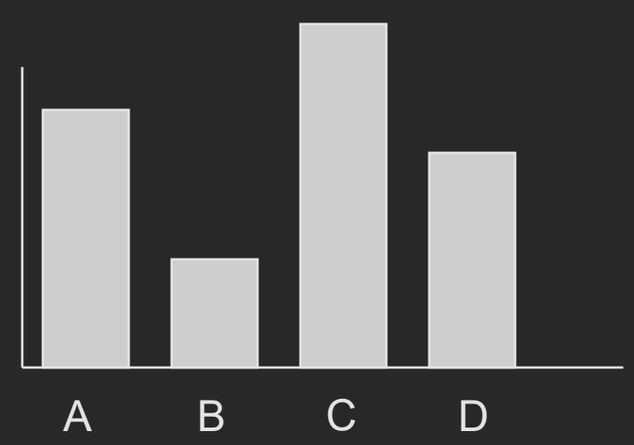
observations

	v1	v2	v3	
A				
B				
C				



Tukey box plot



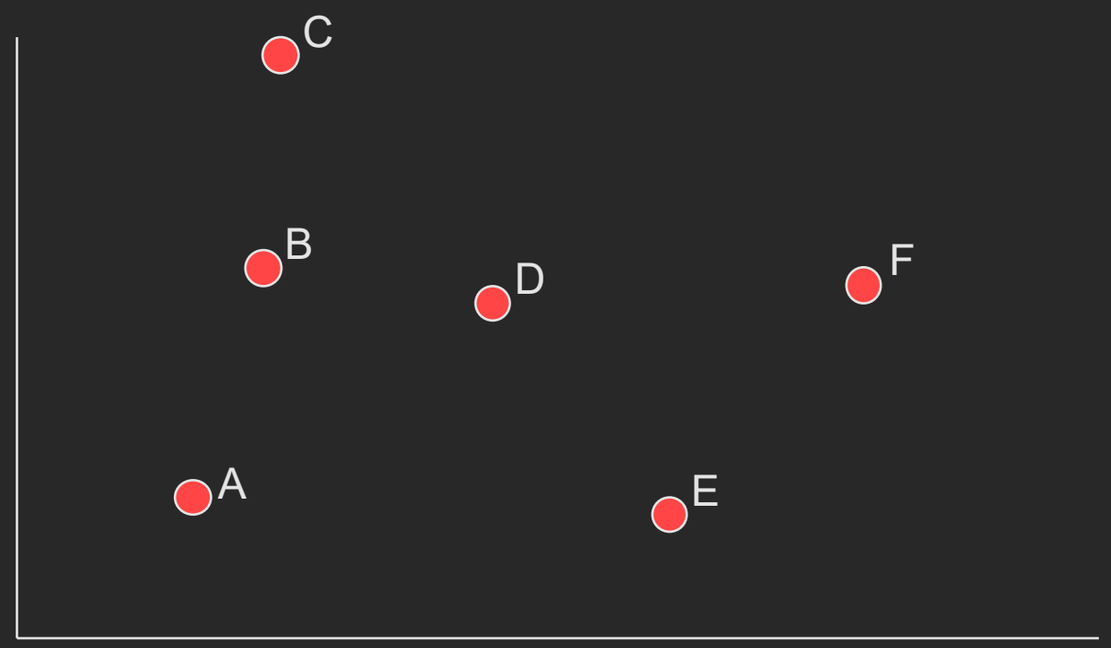


Bivariate data

variables

observations

	v1	v2	v3	
A				
B				
C				



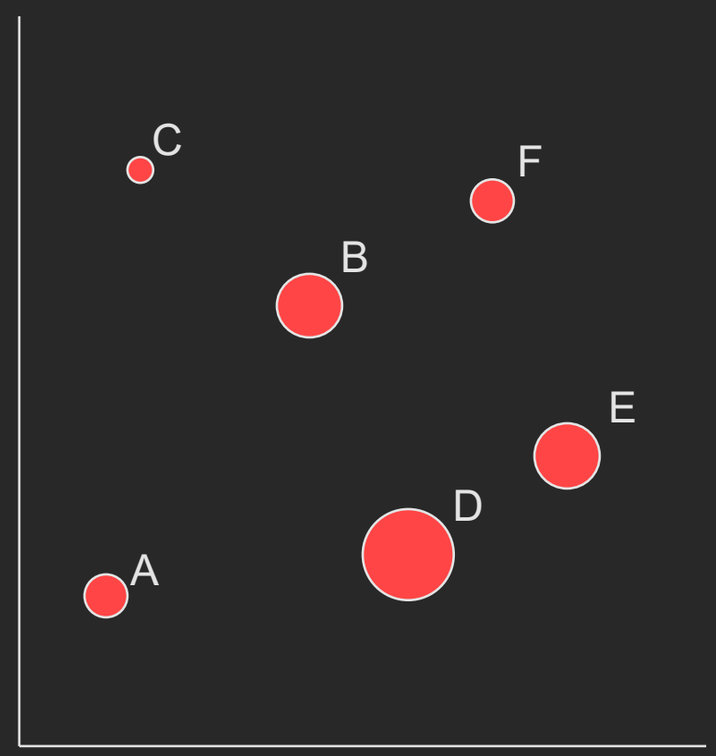
Scatter plot is common

Trivariate data

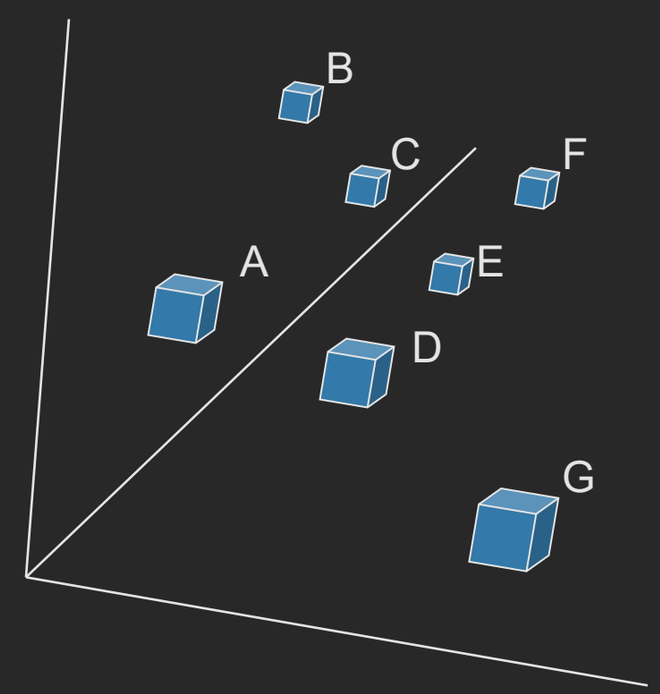
variables

observations

	v1	v2	v3	
A				
B				
C				



3D scatter plot is possible



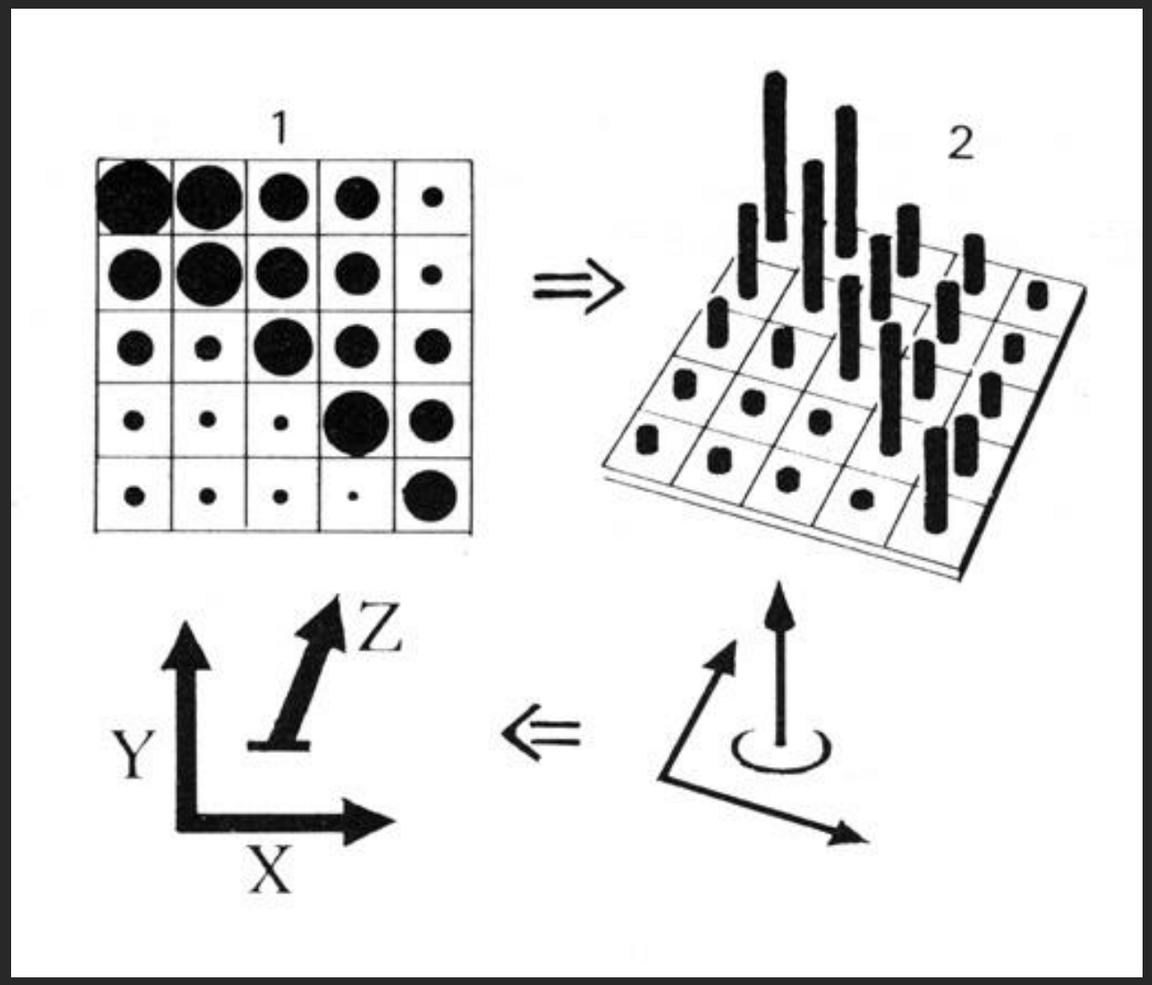
Three variables

Two variables [x,y] can map to points

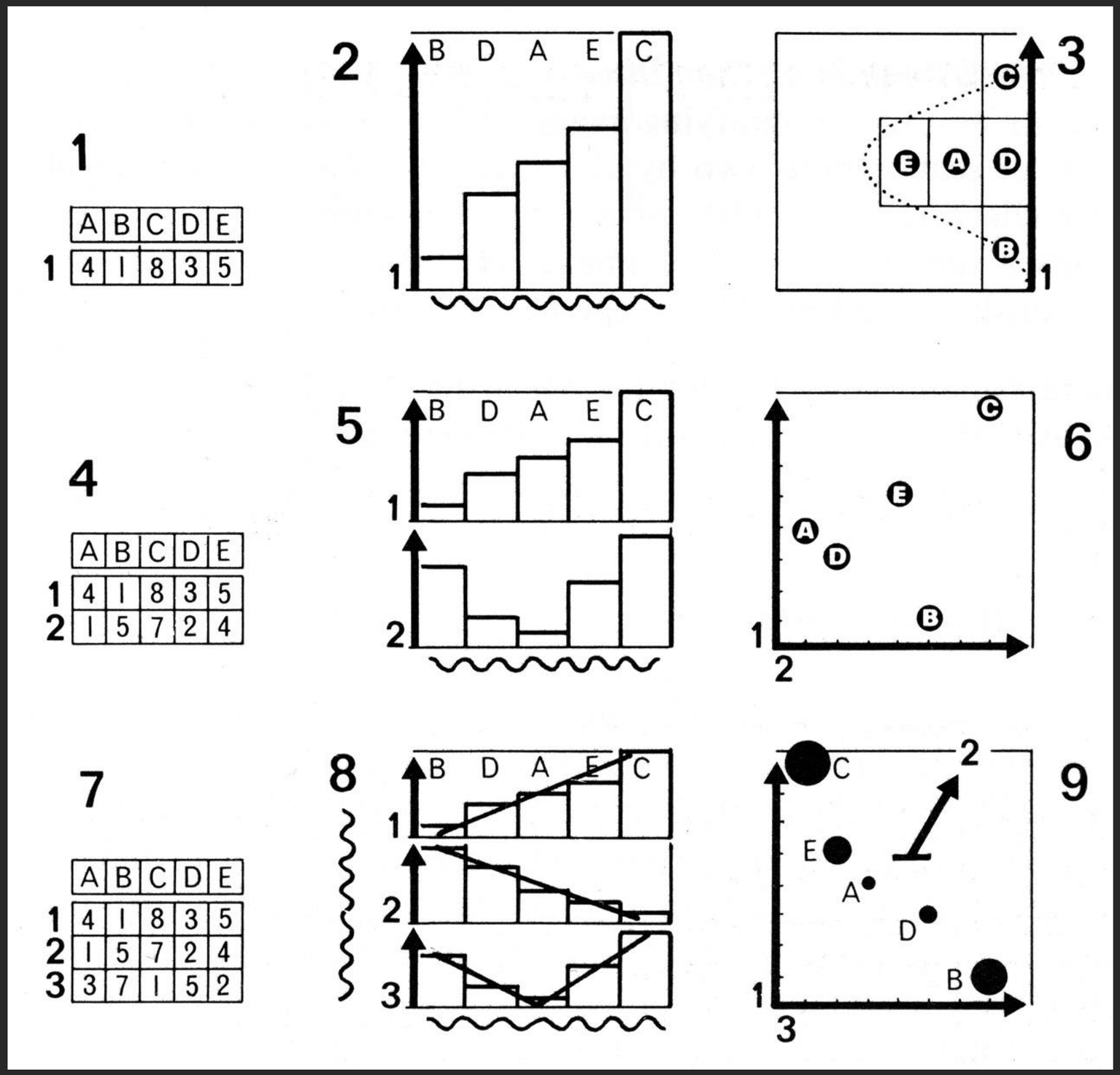
- Scatterplots, maps, …

Third variable [z] must use …

- Color, size, shape, …



Large design space (visual metaphors)



[Bertin, Graphics and Graphic Info. Processing, 1981]

Multidimensional data

How many variables can be depicted in an image?

	v1	v2	v3	v4	v5	v6	v7	…	vN
A									
B									
C									

The big picture

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e.g., find outliers,
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make decision

data

physical type
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abstract type
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visual metaphor

image

visual channel
retinal variables

Data models vs. Conceptual models

Data models: low level descriptions of the data

- Math: Sets with operations on them
- Example: integers with + and $\times$ operators

Conceptual models: mental constructions

- Includes semantics and causality
- Supports reasoning

Examples (data vs. conceptual)

- 1D floats vs. Temperature
- 3D vector of floats vs. Space

Nominal, ordinal and quantitative

N - Nominal (labels)

- Fruits: Apples, oranges, …

O - Ordered

- Quality of meat: Grade A, AA, AAA

Q - Interval (Location of zero arbitrary)

- Dates: Jan, 19, 2006; Location: (LAT 33.98, LONG -118.45)
- Zero is arbitrary. Cannot divide points on scale to get a ratio.
- Only differences (i.e. intervals) may be compared.

Q - Ratio (zero fixed)

- Physical measurement: Length, Mass, Temp, …
- Counts and amounts
- Zero is meaningful. Ratios can be used for comparison.

S. S. Stevens, On the theory of scales of measurements, 1946

Nominal, ordinal and quantitative

N - Nominal (labels)

- Operations: =, ≠

O - Ordered

- Operations: =, ≠, <, >, ≤, ≥

Q - Interval (Location of zero arbitrary)

- Operations: =, ≠, <, >, ≤, ≥, -
- Can measure distances or spans

Q - Ratio (zero fixed)

- Operations: =, ≠, <, >, ≤, ≥, -, ÷
- Can measure ratios or proportions

S. S. Stevens, On the theory of scales of measurements, 1946

From data model to N,O,Q data type

Data model

- 32.5, 54.0, -17.3, …
- floats

Conceptual model

- Temperature

Data type

- Burned vs. Not burned (N)
- Hot, warm, cold (O)
- Continuous range of values (Q)

How do these ways of thinking about data help us choose visual encodings?

Bertin's "Levels of Organization"

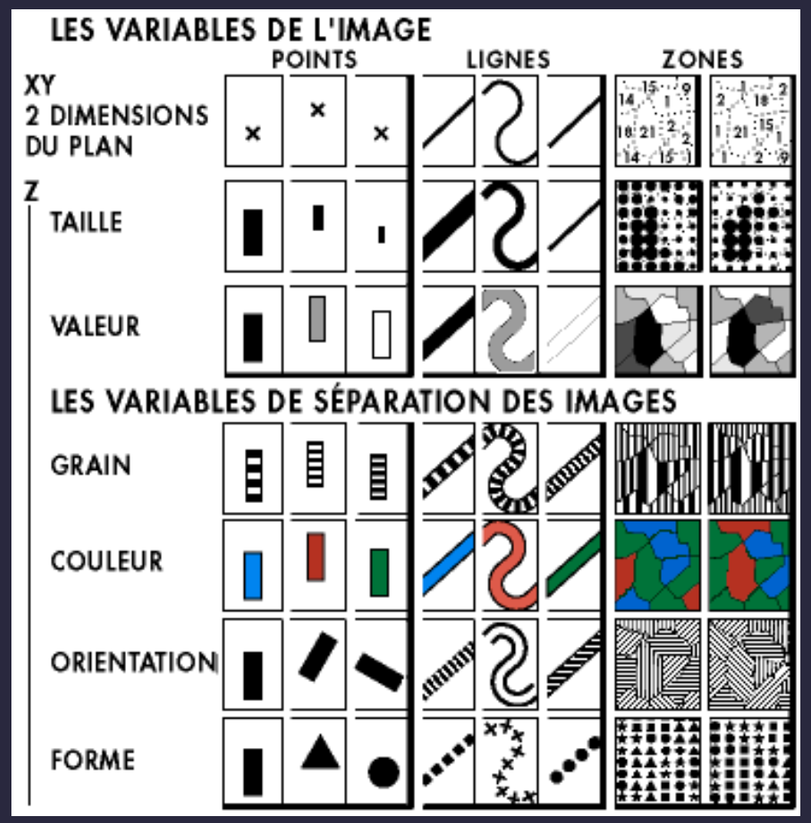
Position	N	O	Q
Size	N	O	Q
Value	N	O	Q
Texture	N	o	
Color	N		
Orientation	N		
Shape	N		

N Nominal

O Ordered

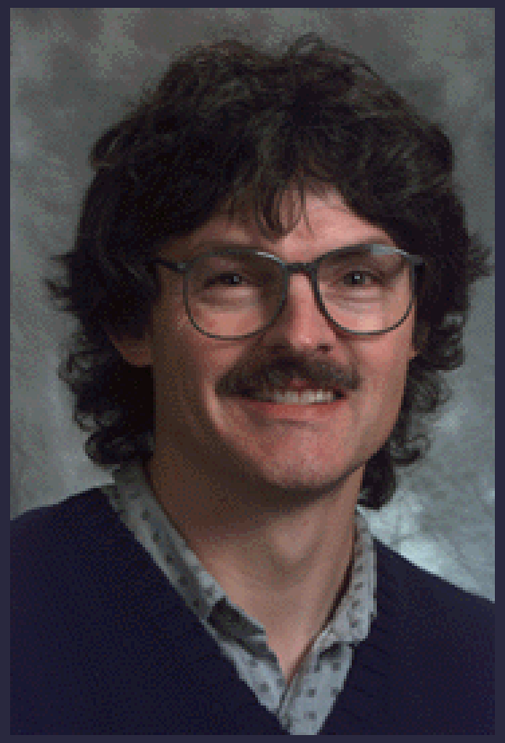
Q Quantitative

Note: Q < O < N



Automated design

Jock Mackinlay's APT 86



Combinatorics of encodings

Challenge:

Assume 8 visual encodings and n data attributes

Pick the best encoding from the exponential number of possibilities $(n+1)^8$

Principle of Consistency:

The properties of the image (visual variables) should match the properties of the data

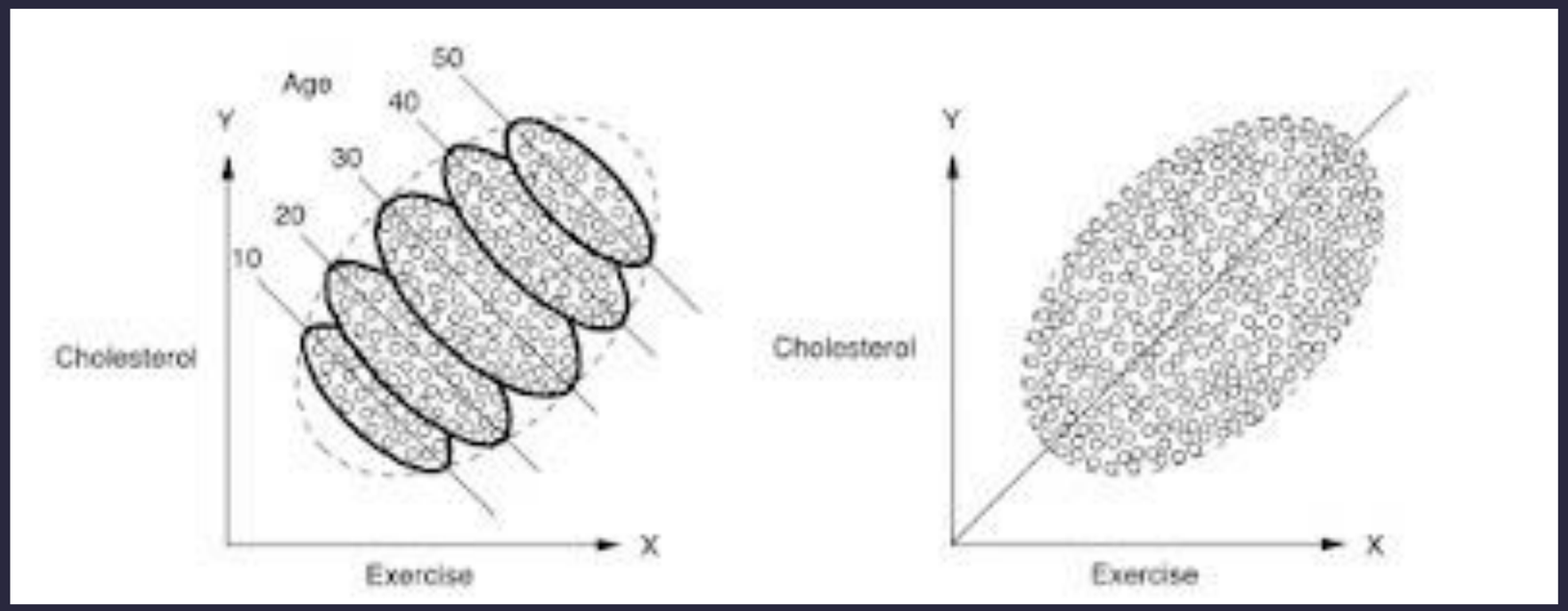
Principle of Importance Ordering:

Encode the most important information in the most effective way

Mackinlay's expressiveness criteria

Expressiveness

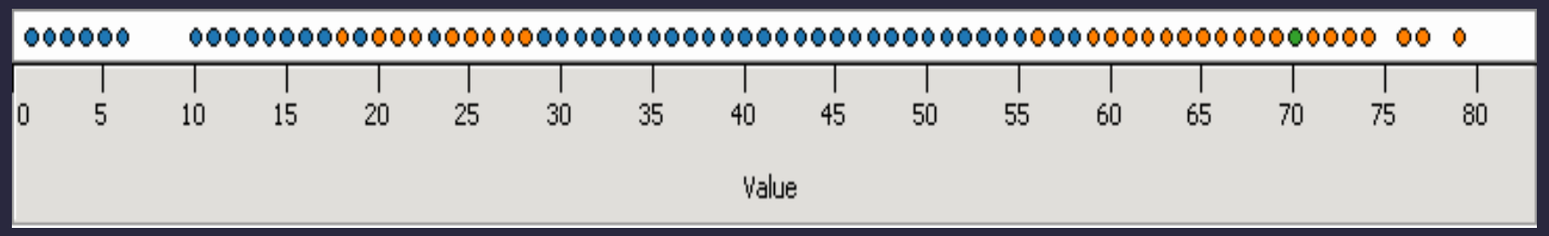
A set of facts is expressible in a visual language if the sentences (i.e. the visualizations) in the language express *all* the facts in the set of data, and *only* the facts in the data.

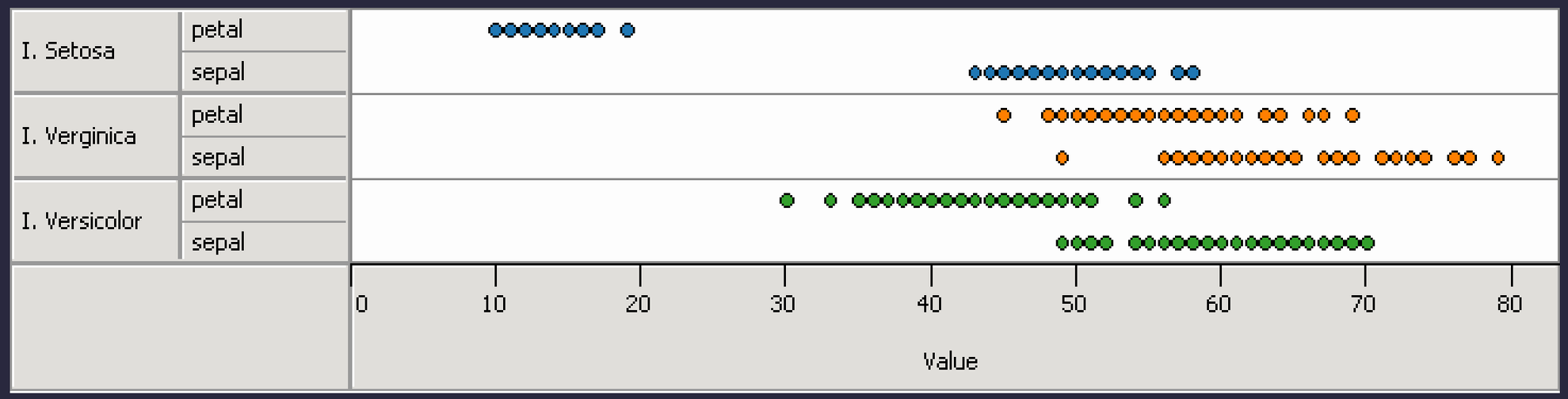


Pearl (2018) The Book of Why.

Cannot express the facts

A one-to-many (1 → N) relation cannot be expressed in a single horizontal dot plot because multiple tuples are mapped to the same position





Expresses facts not in the data

A length is interpreted as a quantitative value;

∴ Length of bar says something untrue about N data

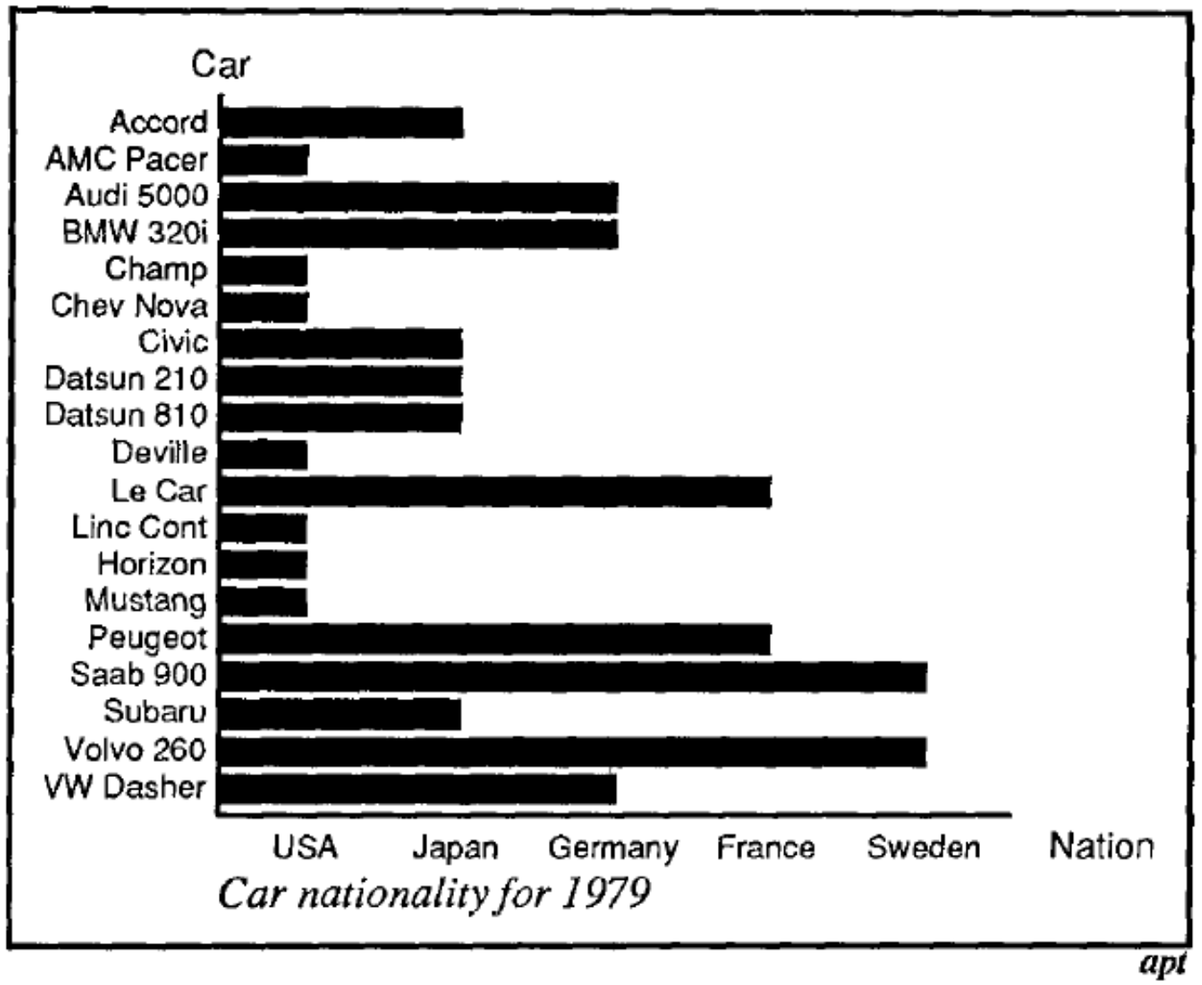


Fig. 11. Incorrect use of a bar chart for the *Nation* relation. The lengths of the bars suggest an ordering on the vertical axis, as if the USA cars were longer or better than the other cars, which is not true for the *Nation* relation.

[Mackinlay, APT, 1986]

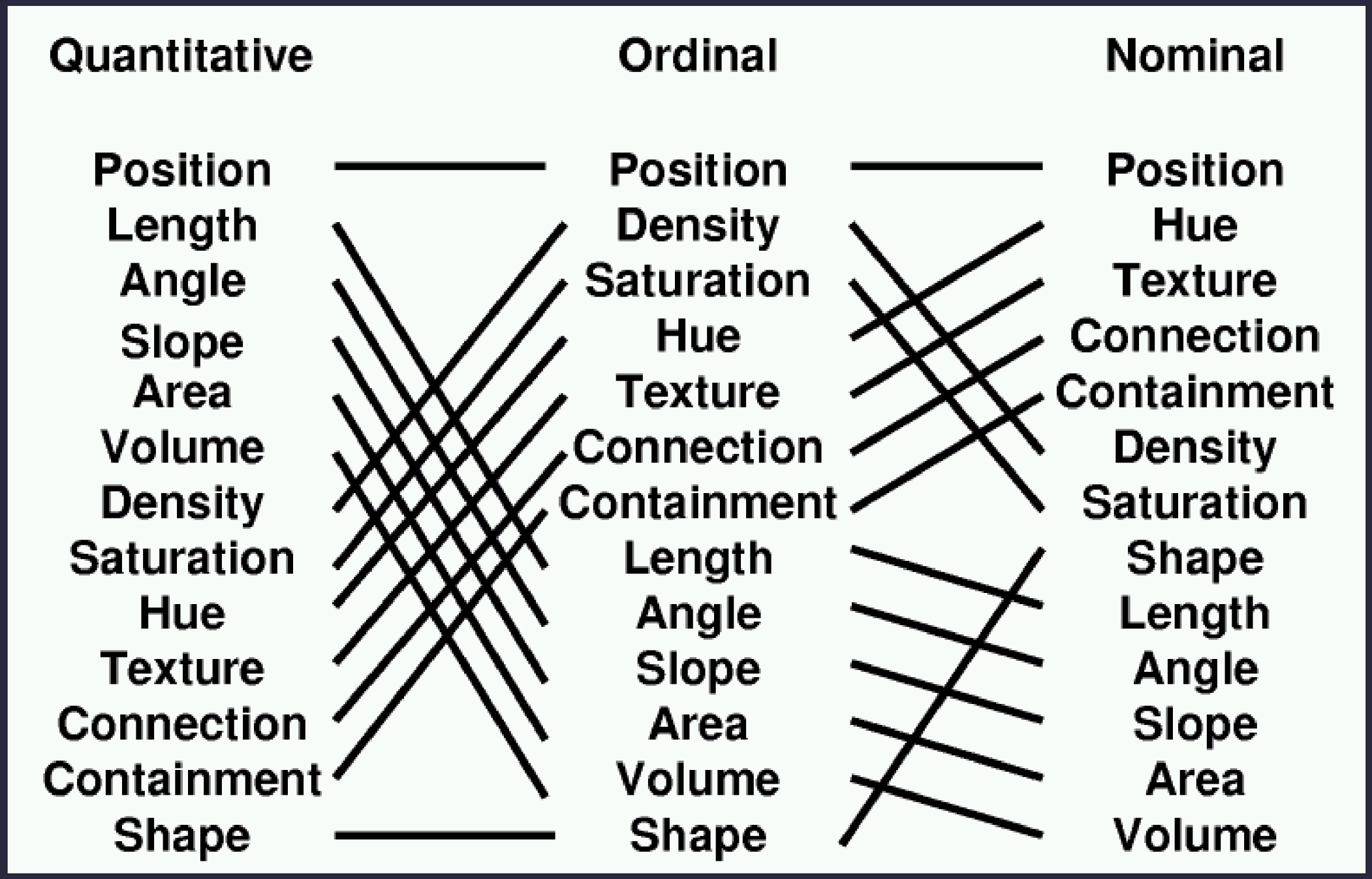
Mackinlay's effectiveness criteria

Effectiveness

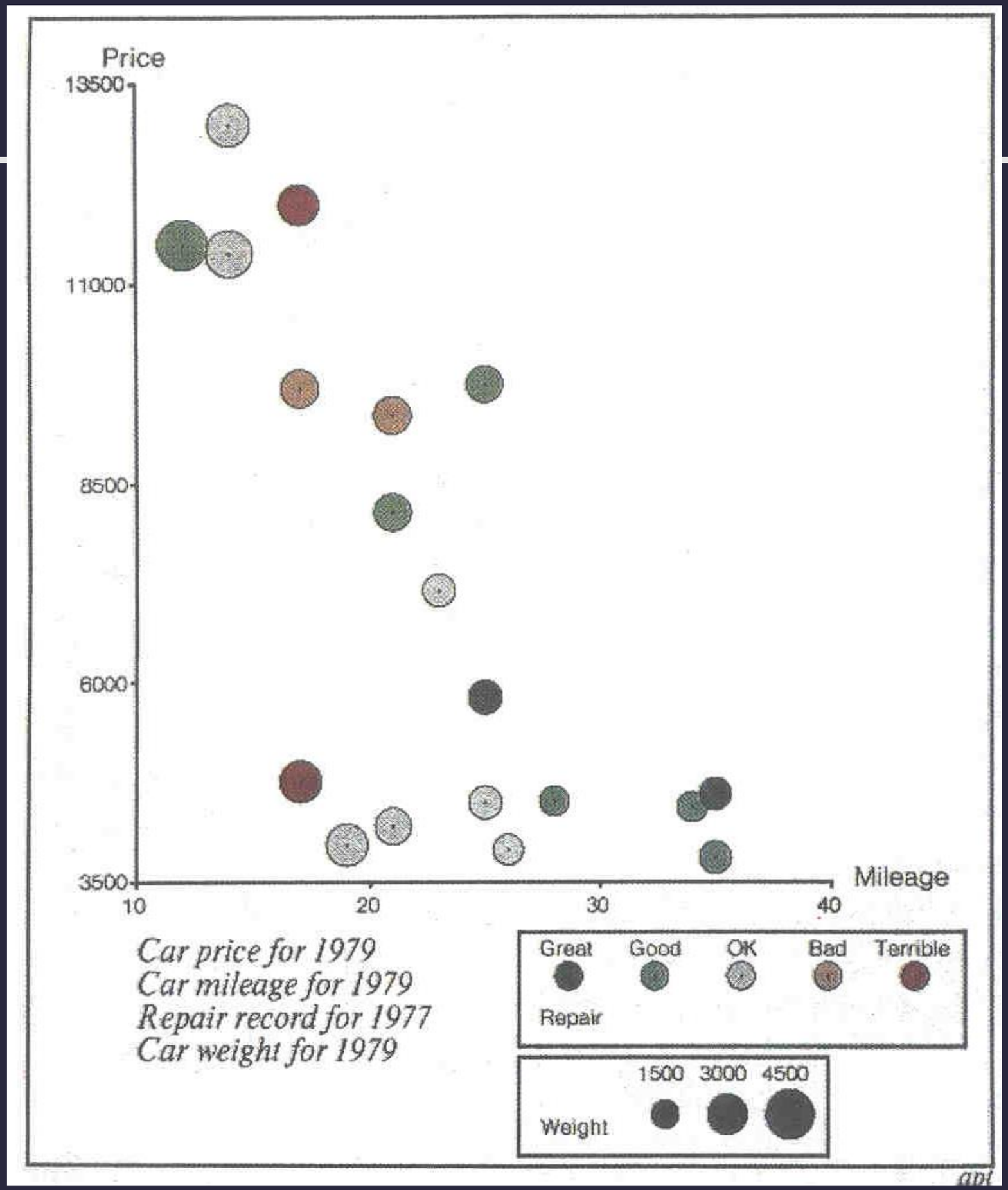
A visualization is more effective than another visualization if the information conveyed by one visualization is more readily *perceived* than the information in the other visualization.

Subject of perception lecture

Mackinlay's ranking



Conjectured *effectiveness* of the encoding



[Mackinlay, APT, 1986]

Summary

Formal specification

- Data model
- Image model (Grammar of graphics)
- Encodings mapping data to image

Choose expressive and effective encodings

- Must check for violations of principles
- Requires knowledge of graphical conventions, mathematical properties of data, empirical research on perception and more
- Automated approaches like APT are promising but not completely reliable

Conventional wisdom

“Show data variation, not design variation”

-Tufte

Tufte: Data-ink ratio

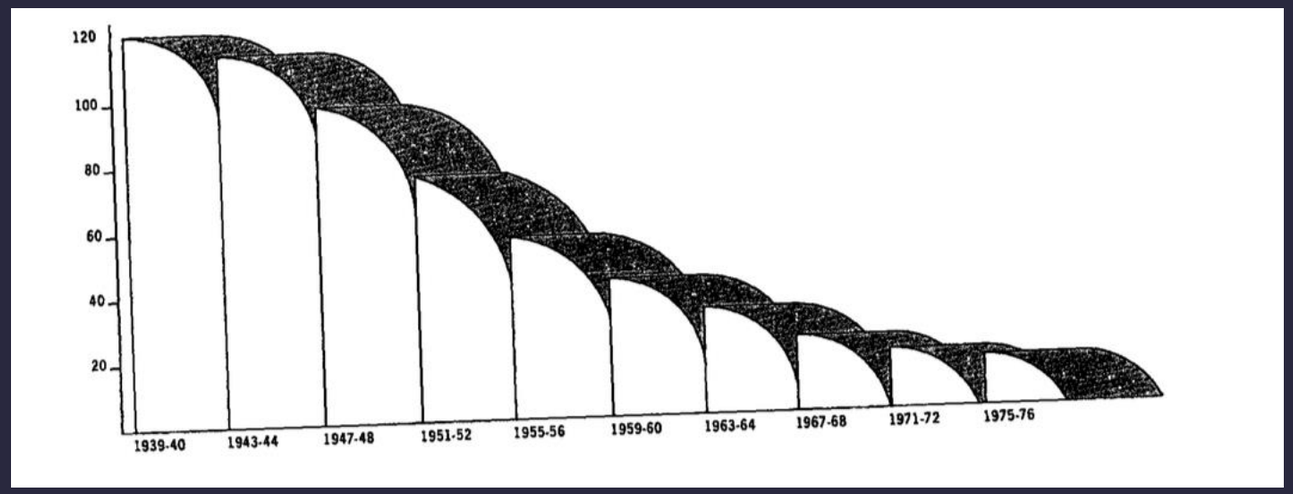
Data-Ink

A large share of ink on a graphic should present data-information, the ink changing as the data change. *Data-ink* is the non-erasable core of a graphic, the non-redundant ink arranged in response to variation in the numbers represented. Then,

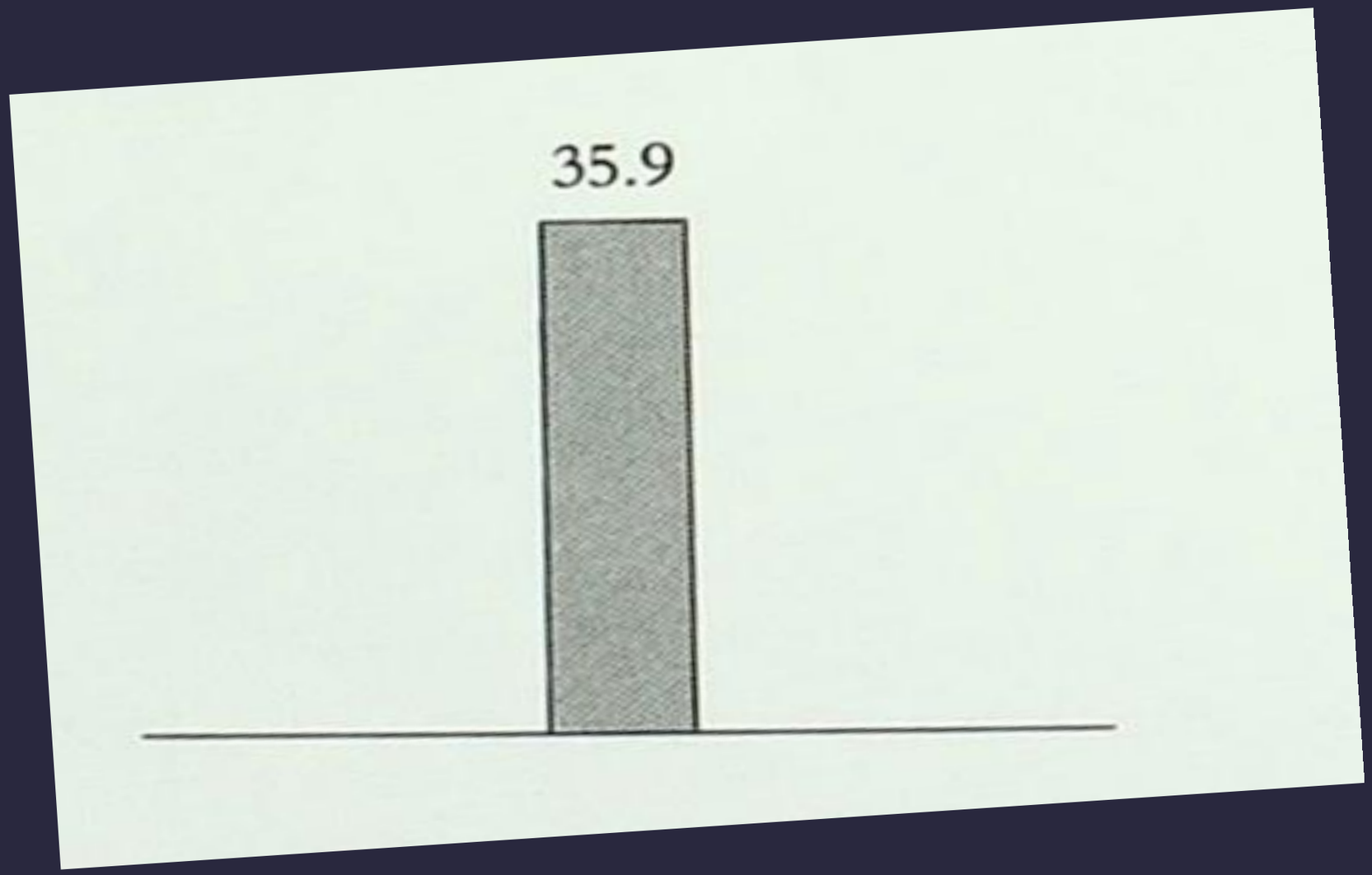
$$\text{Data-ink ratio} = \frac{\text{data-ink}}{\text{total ink used to print the graphic}}$$

= proportion of a graphic's ink devoted to the non-redundant display of data-information

= 1.0 – proportion of a graphic that can be erased without loss of data-information.

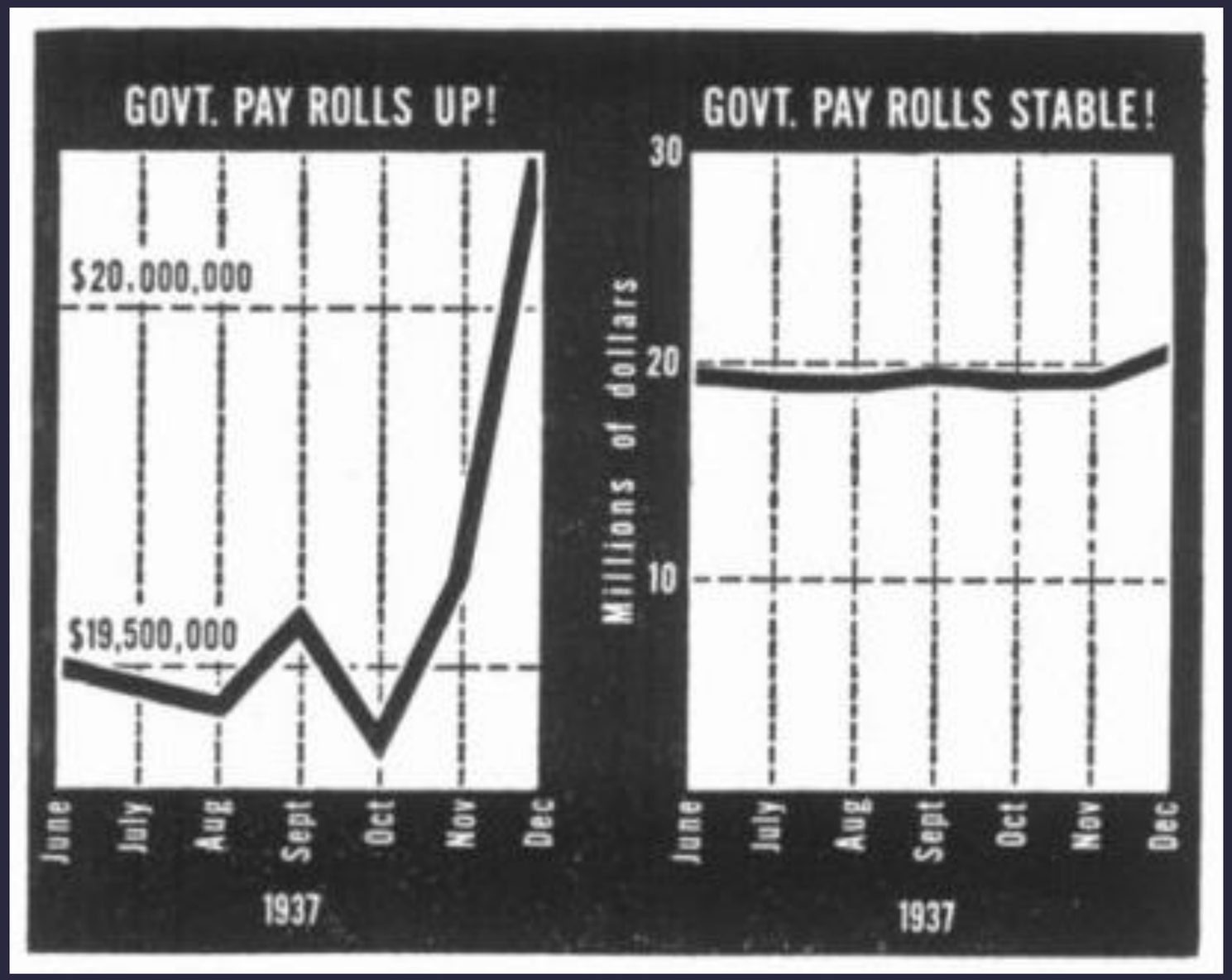


Tufte: Data-ink ratio



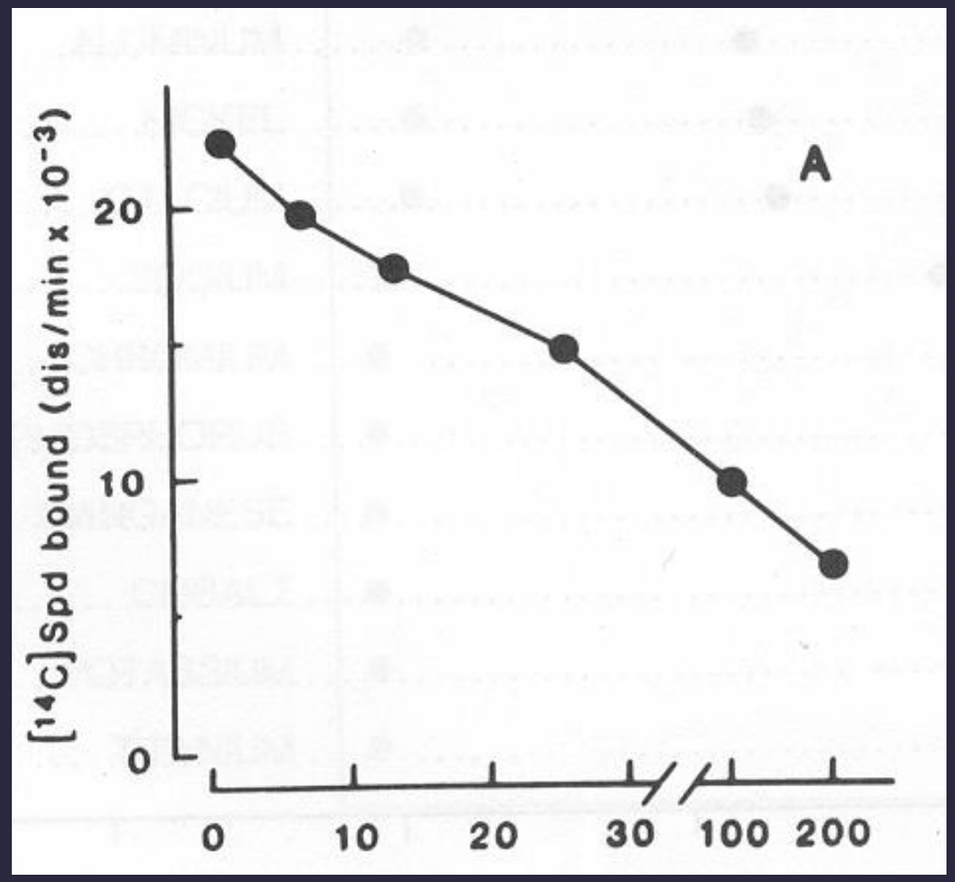
Effective use of space

Which graph is better?

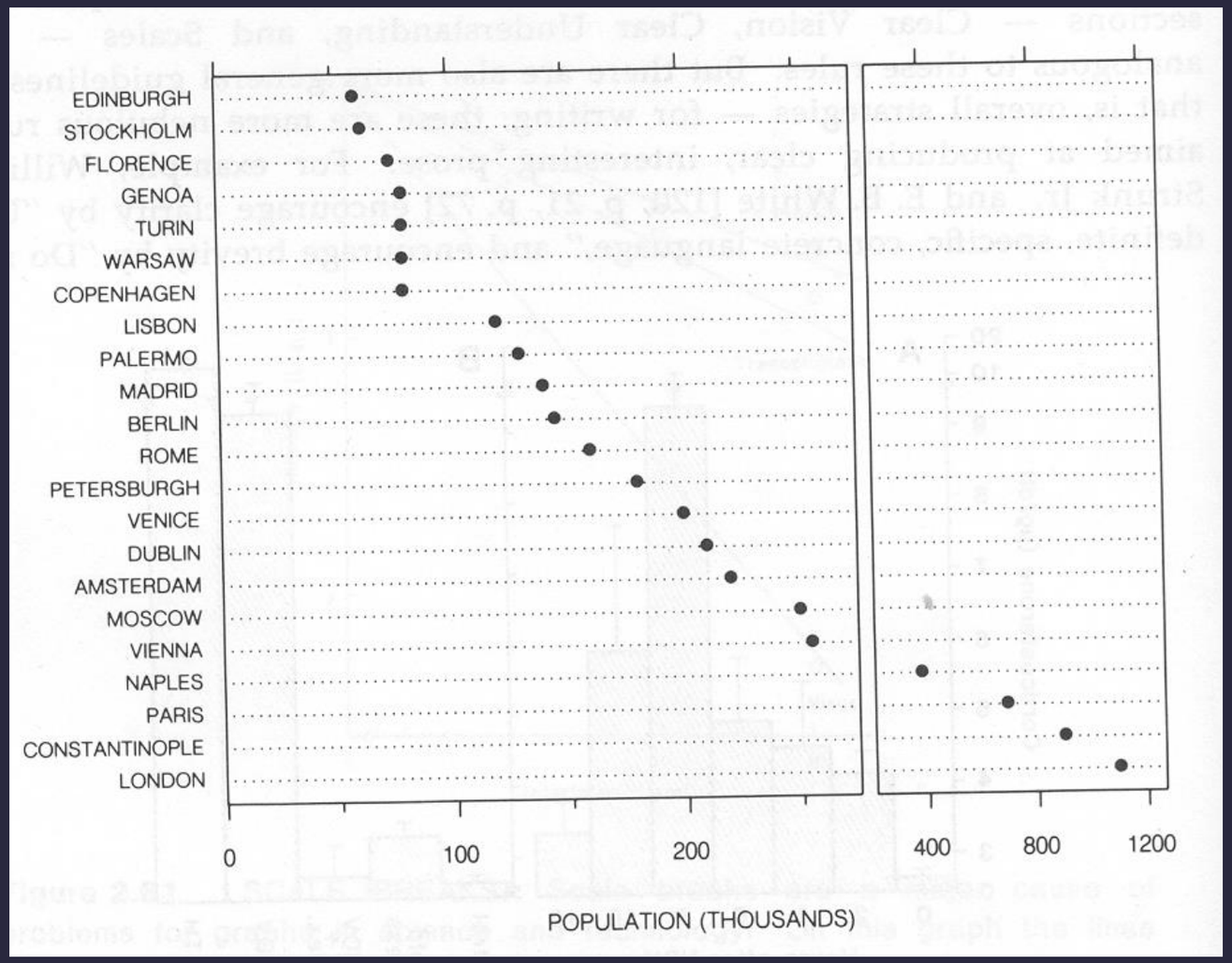


Government payrolls in 1937 [Huff 93]

Clearly mark scale breaks

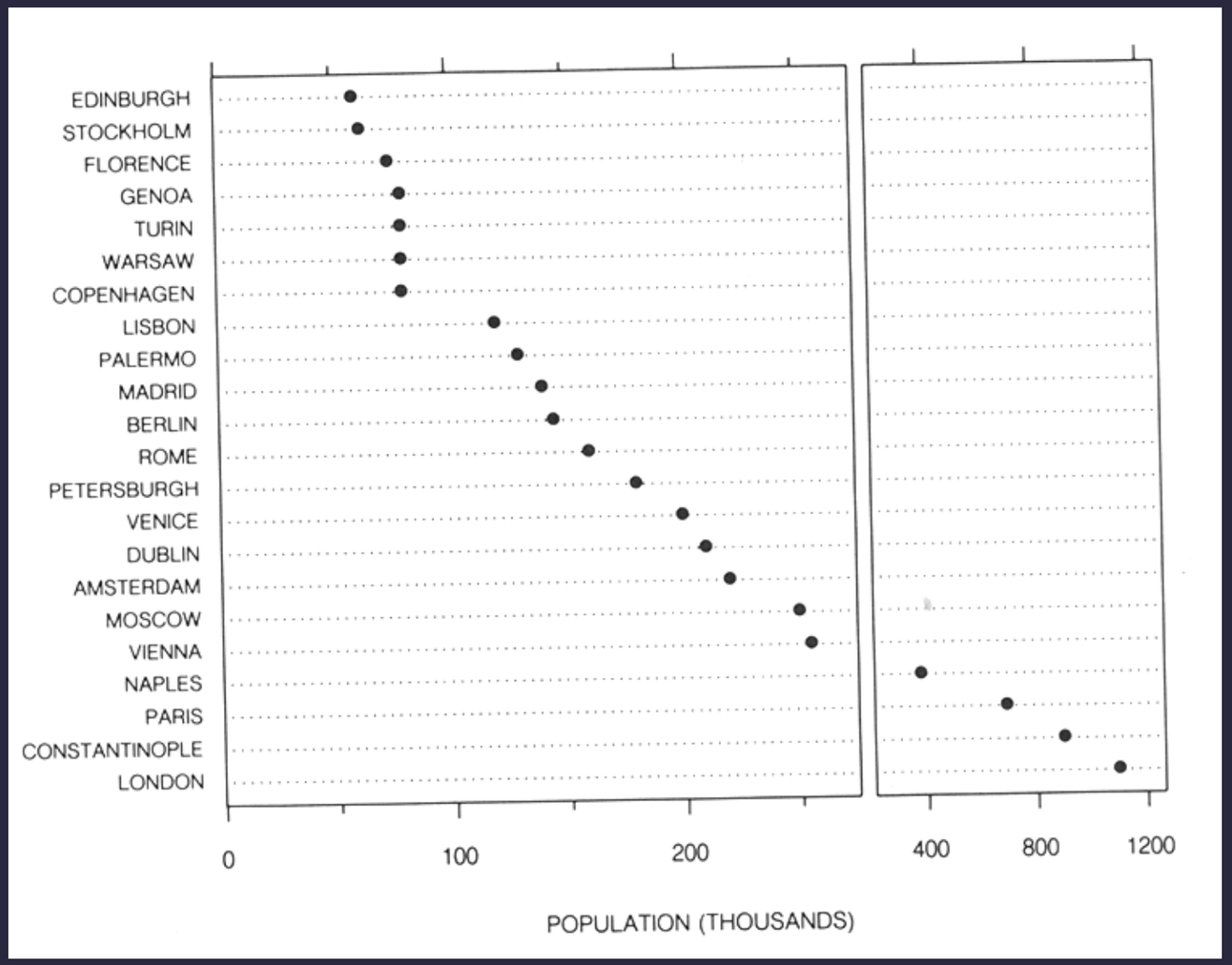


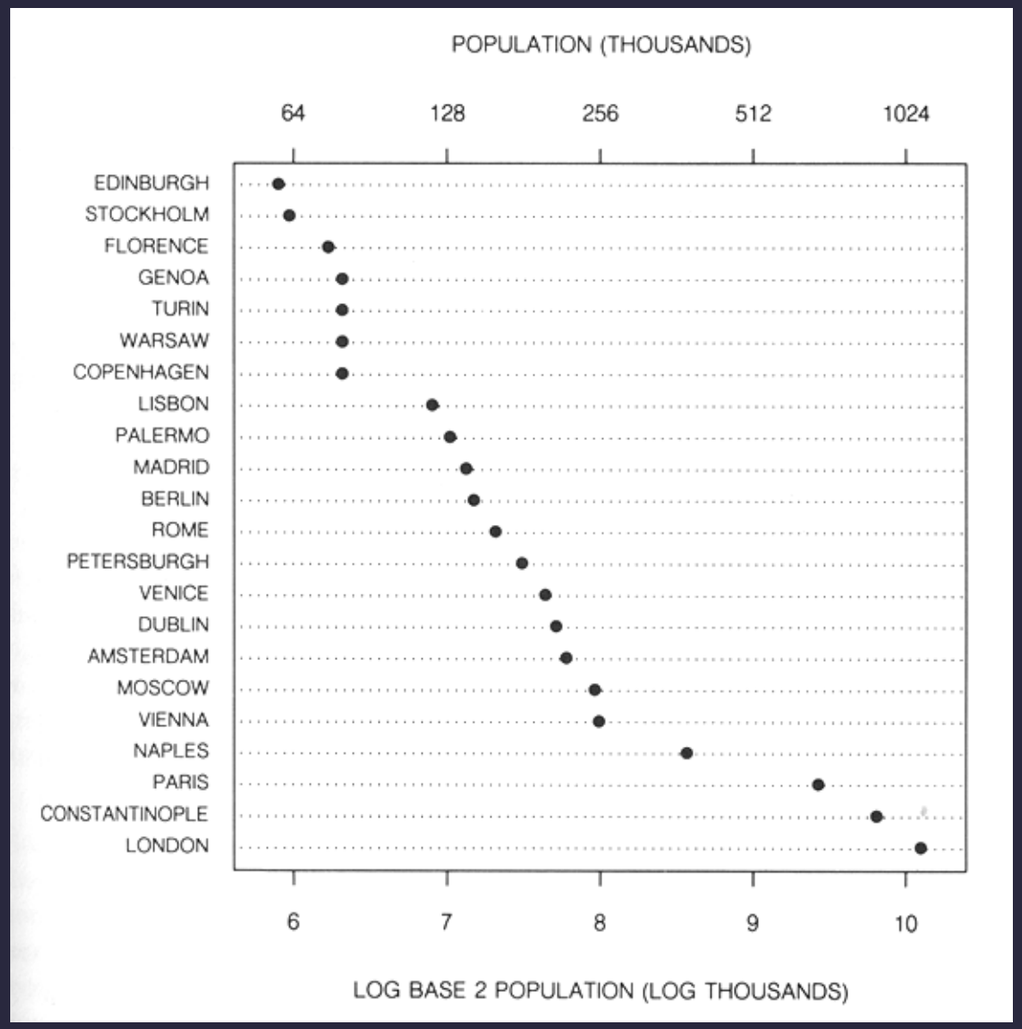
Poor scale break
[Cleveland 85]



Well marked scale break
[Cleveland 85]

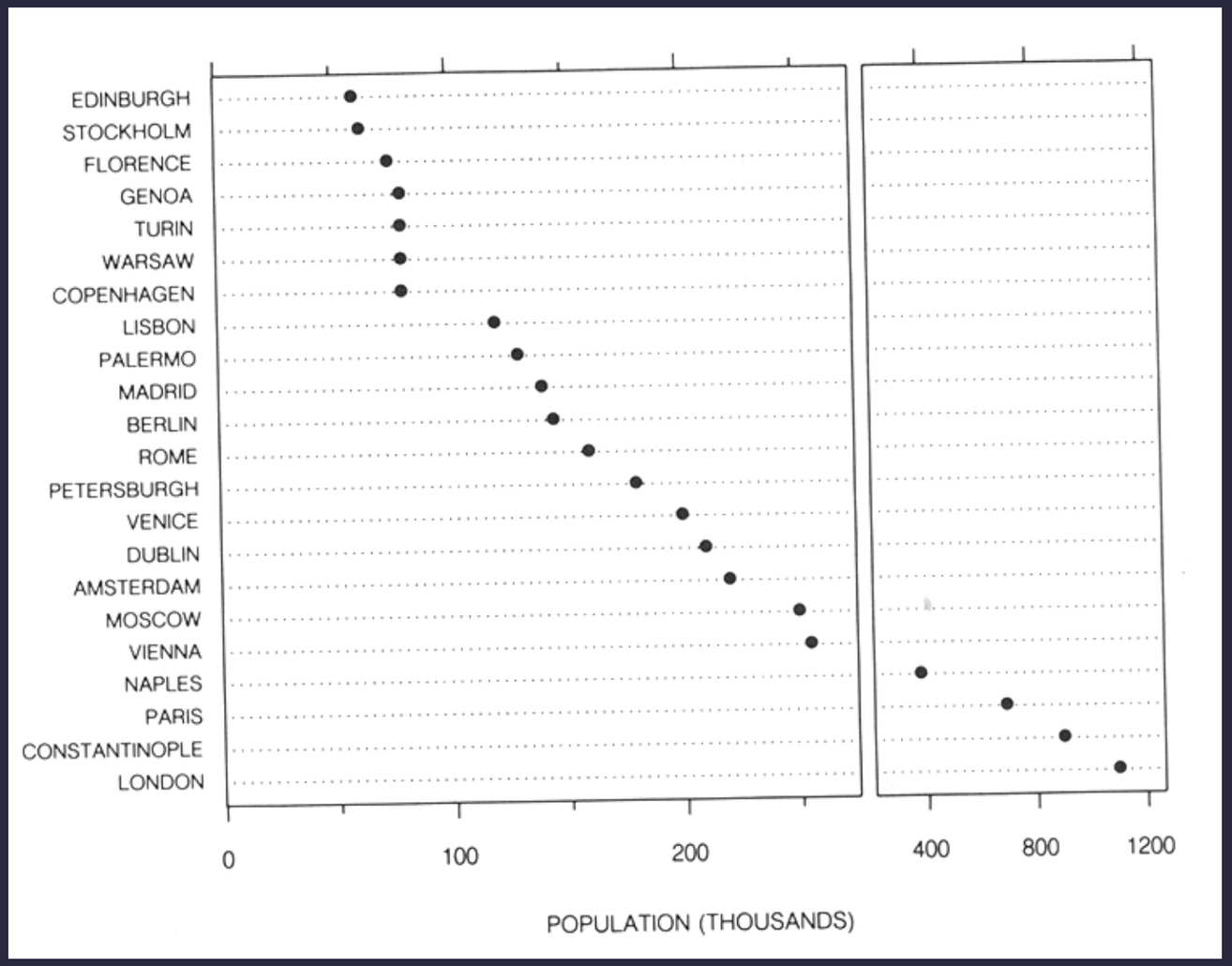
Scale break vs. Log scale

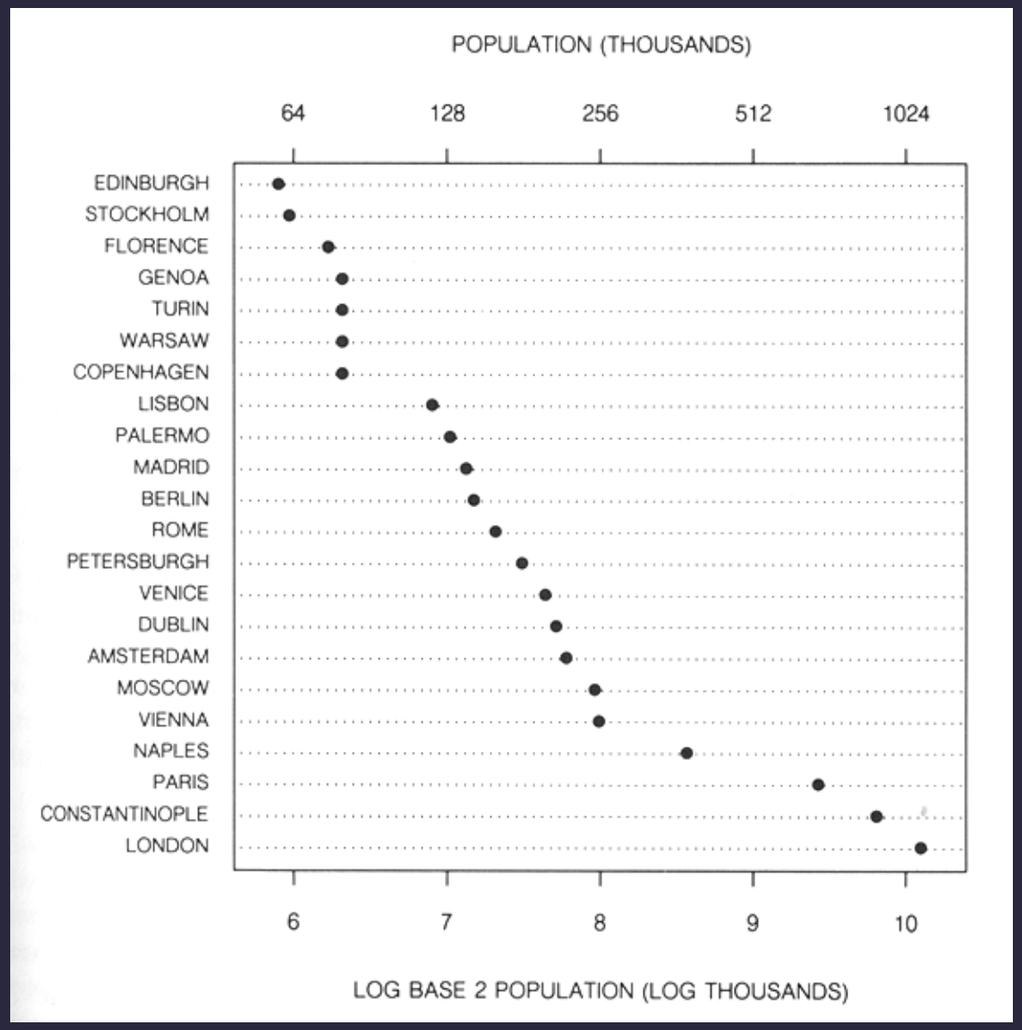




[Cleveland 85]

Scale break vs. Log scale



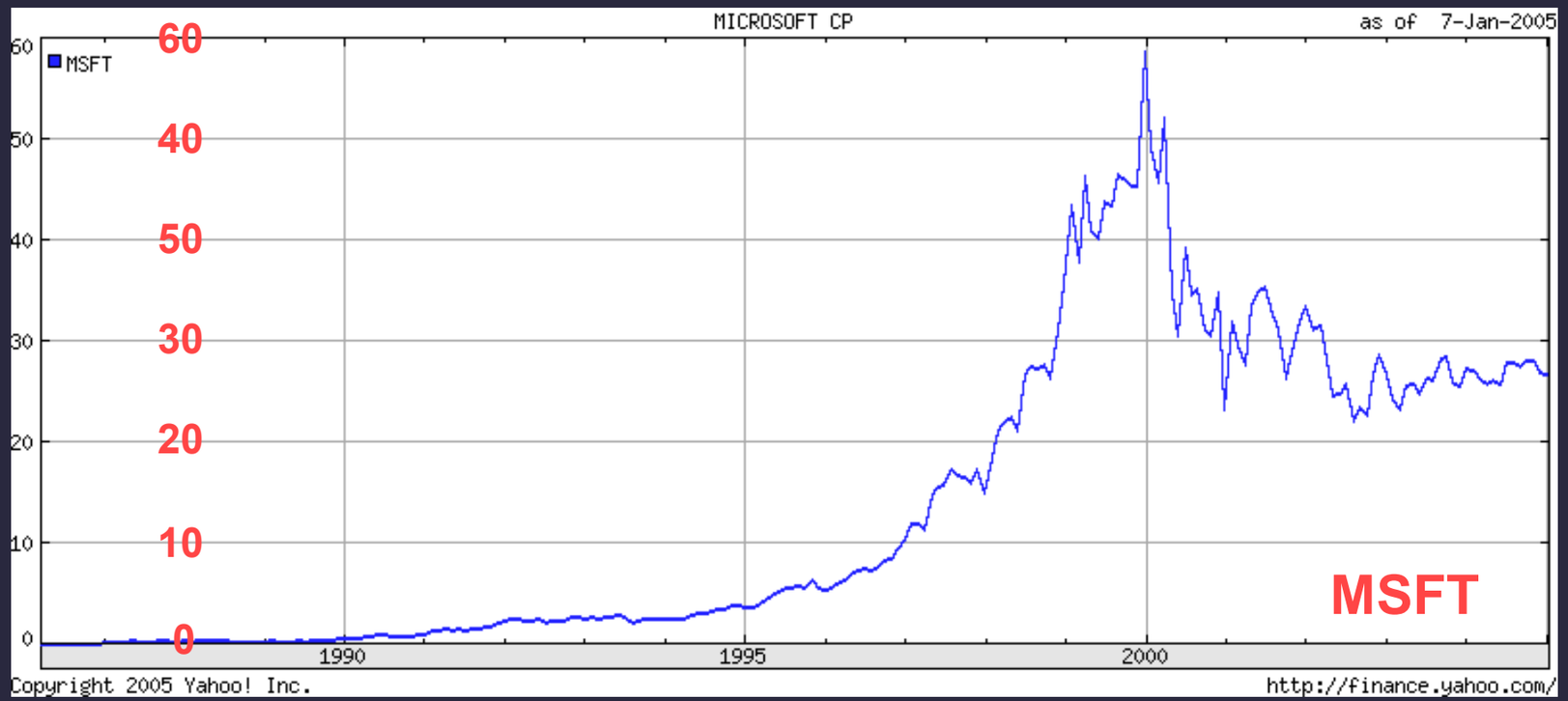


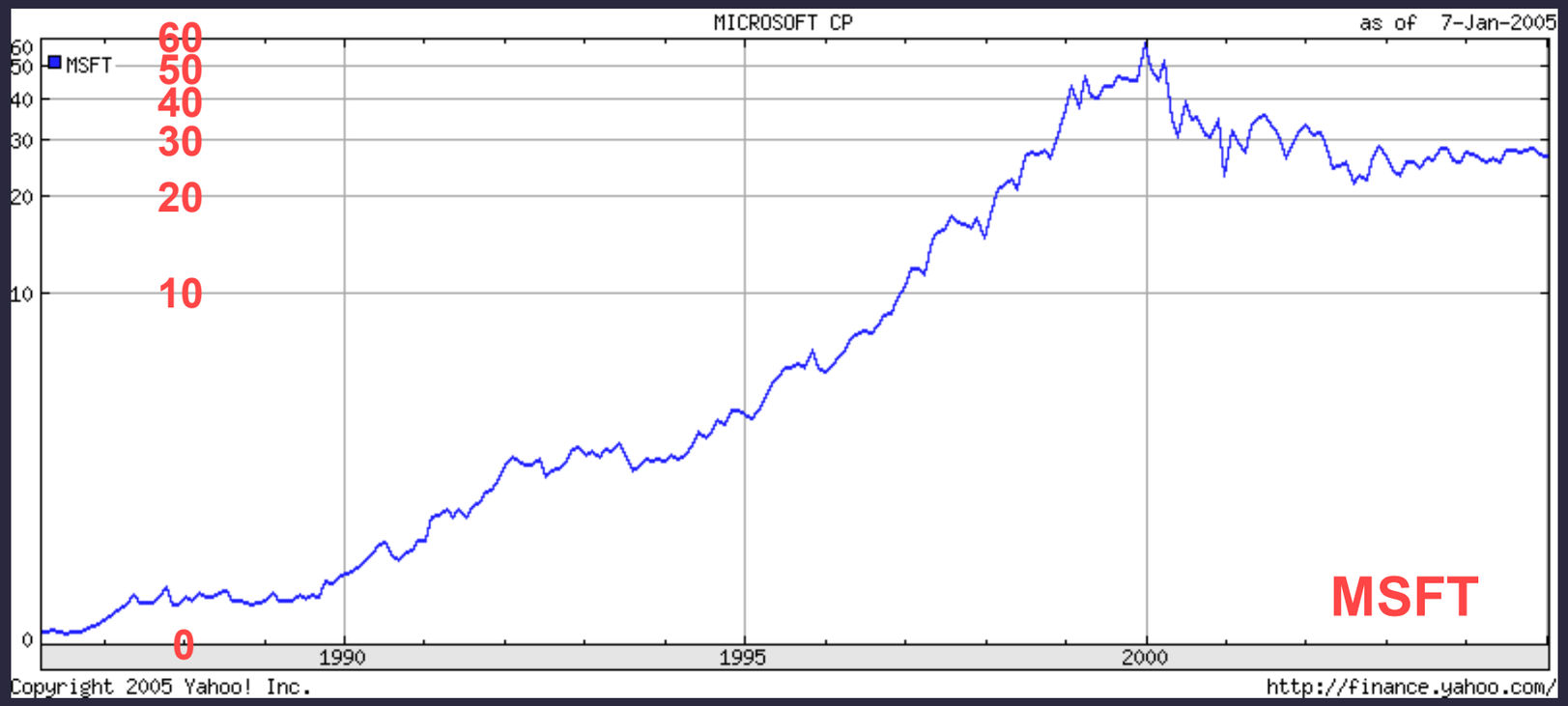
[Cleveland 85]

Both increase visual resolution

- Log scale - easy comparisons of all data
- Scale break – more difficult to compare across break

Linear scale vs. Log scale

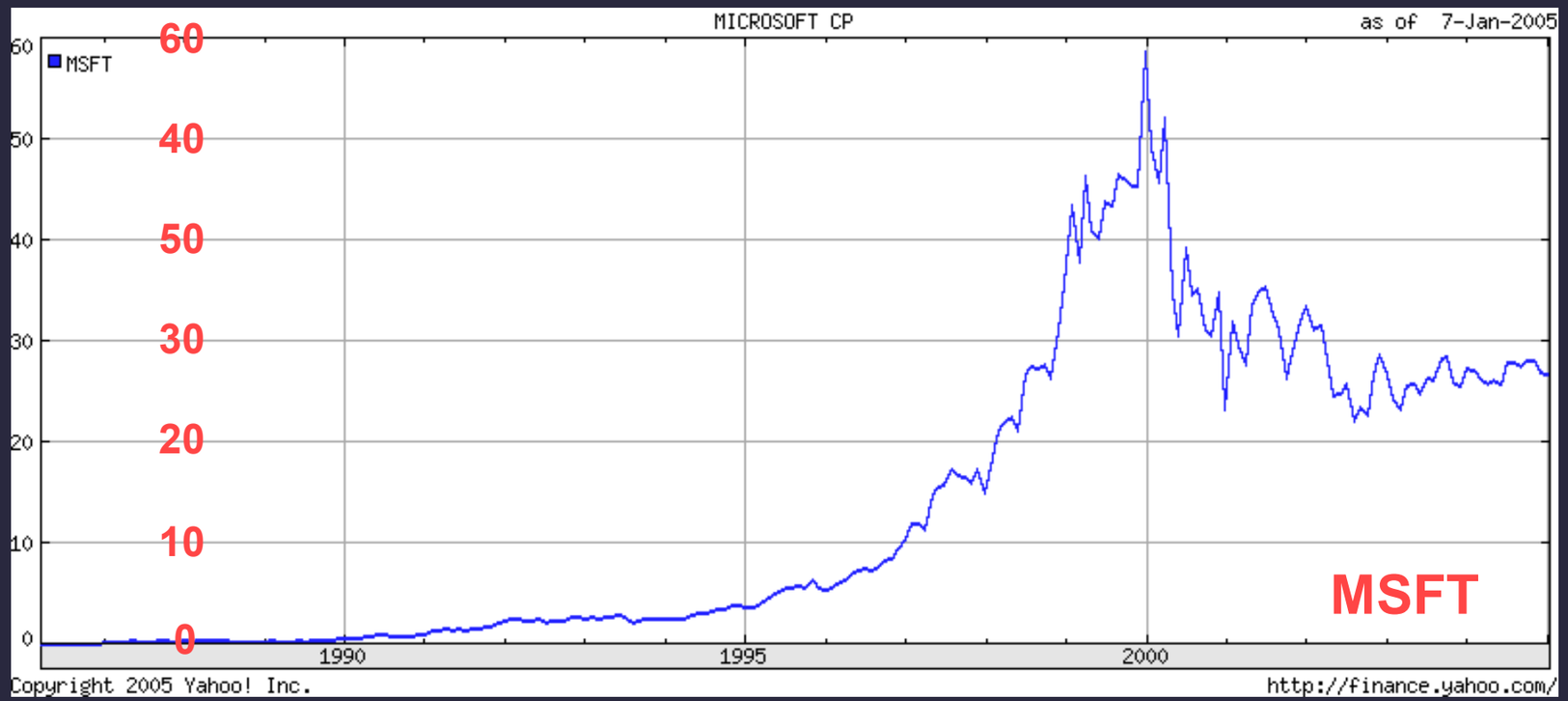




Linear scale vs. Log scale

Linear scale

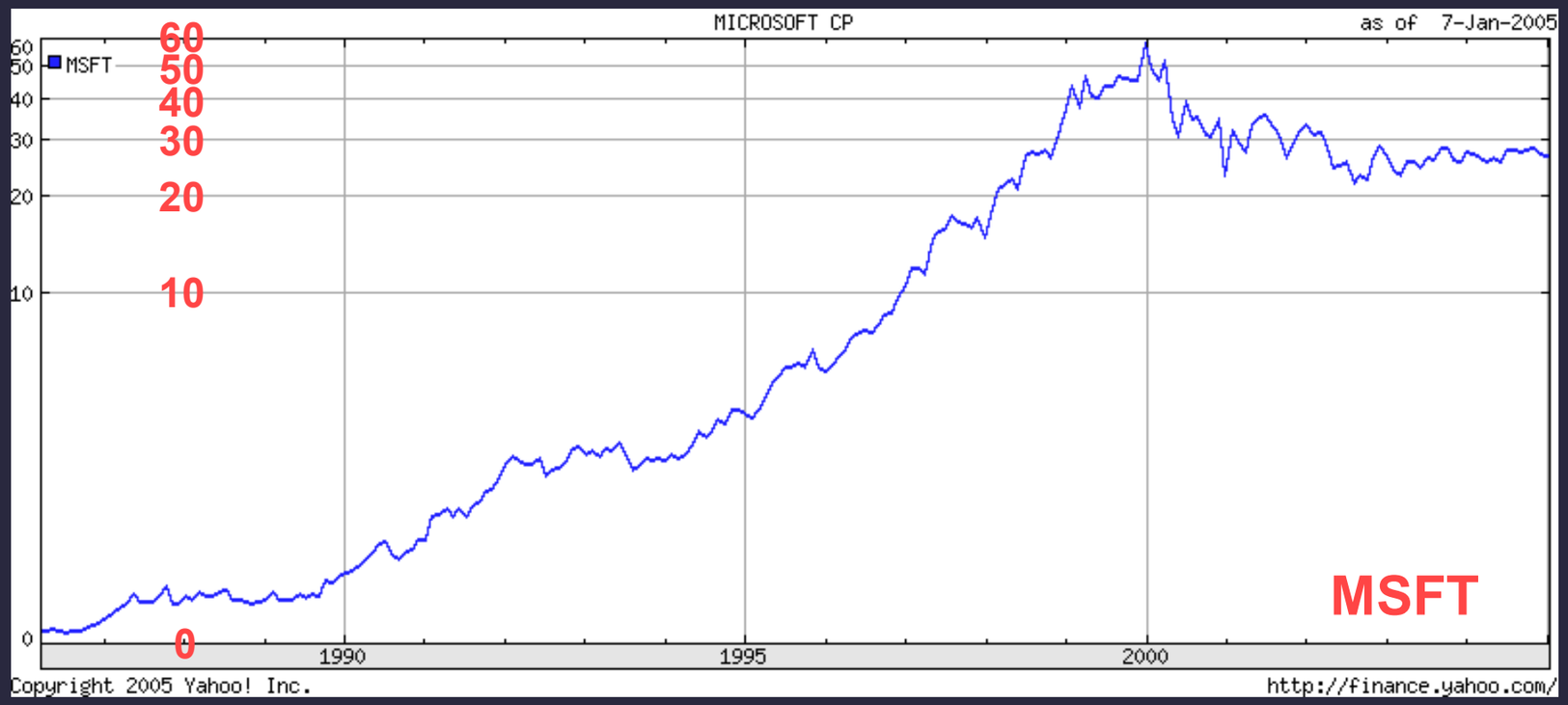
- Absolute change



Log scale

- Small fluctuations
- Percent change

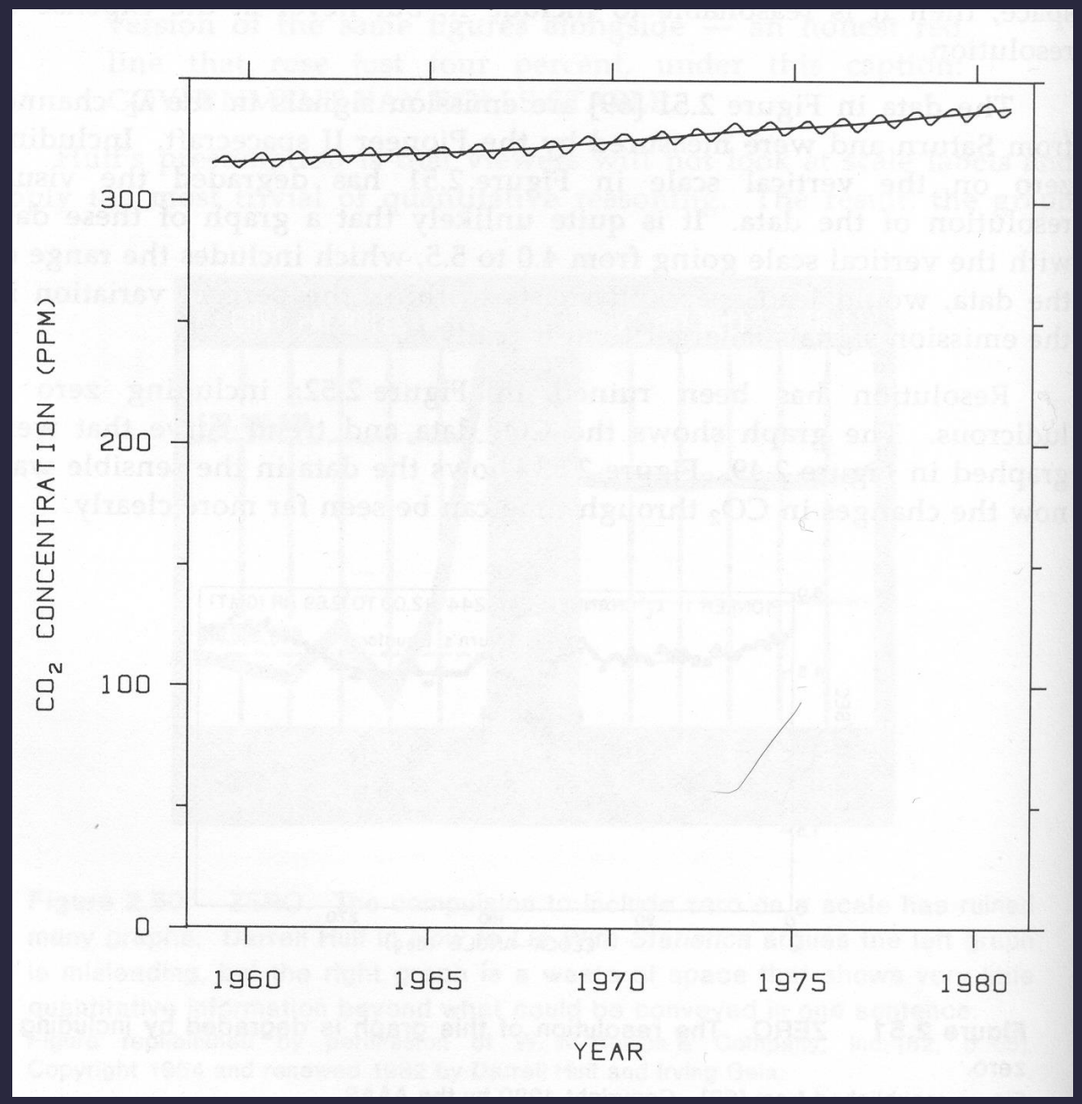
  d(10,20) = d(30,60)

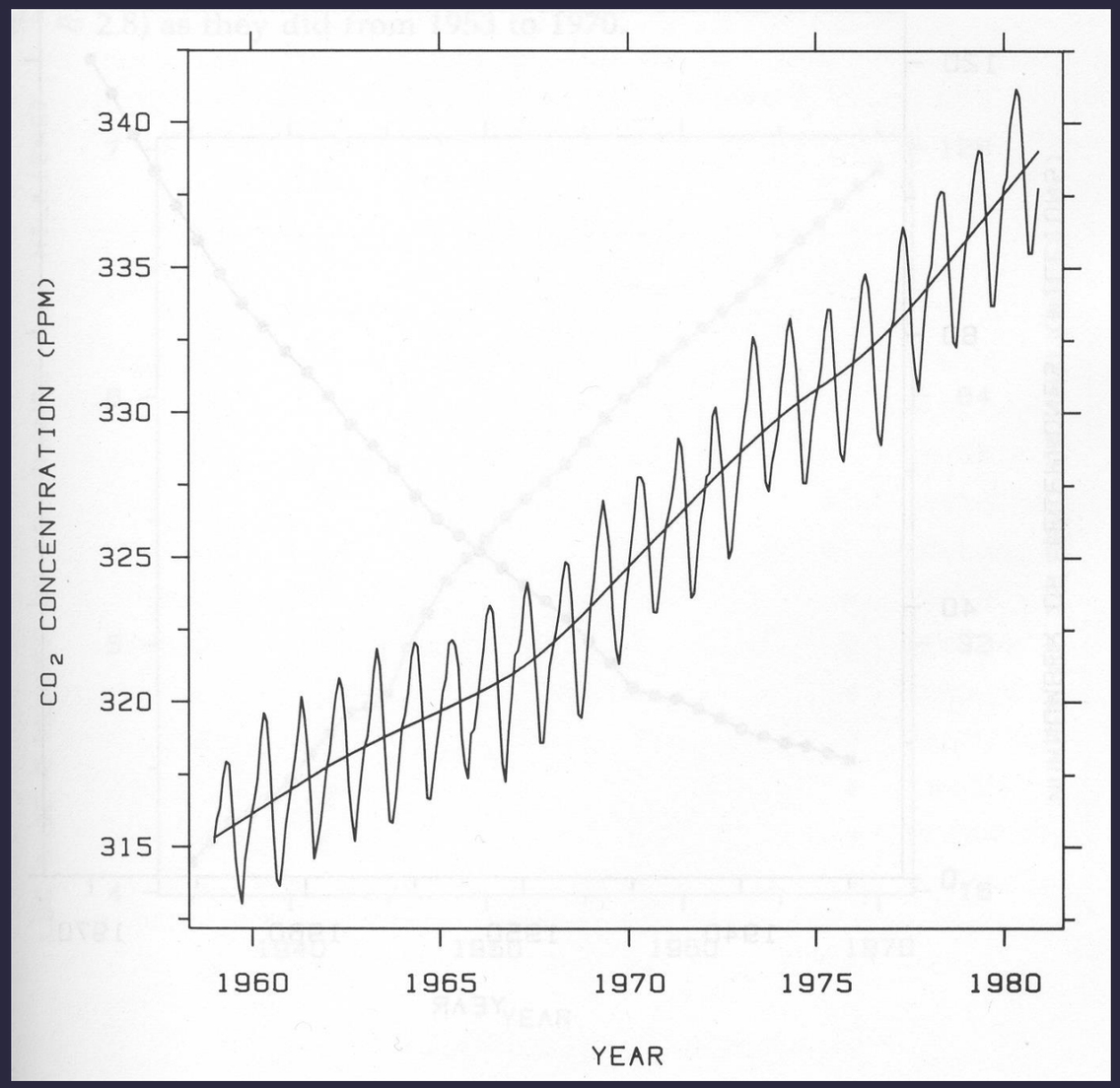


Aspect ratio

Fill space with data

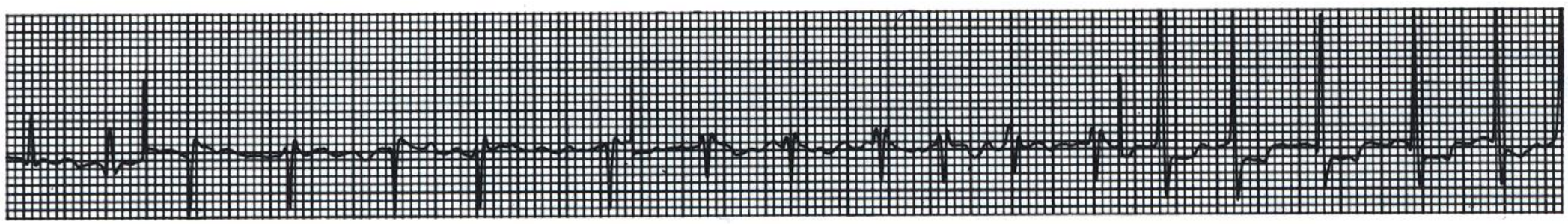
Show zero only if you have a good reason



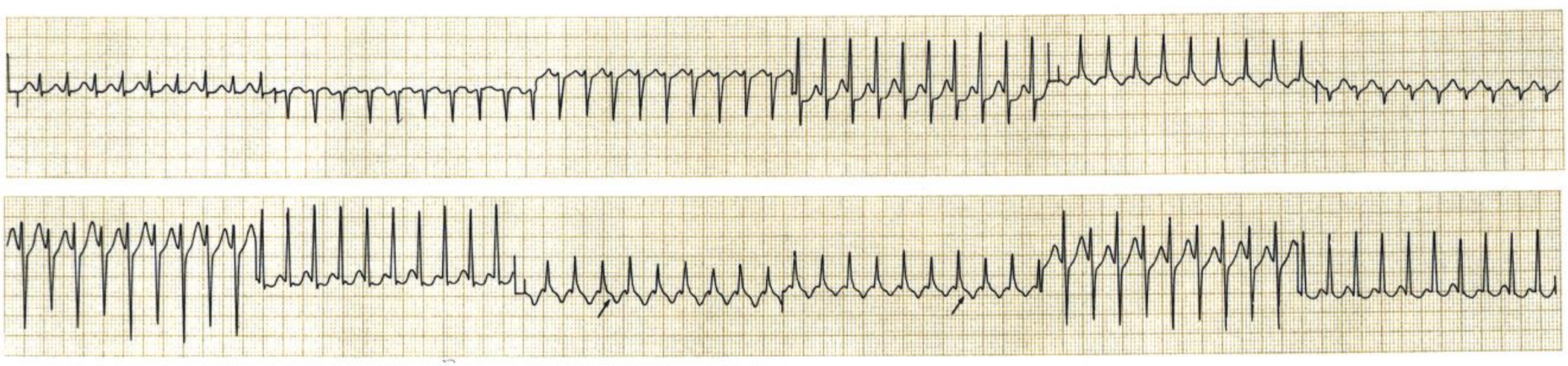


Yearly CO2 concentrations [Cleveland 85]

Gridlines

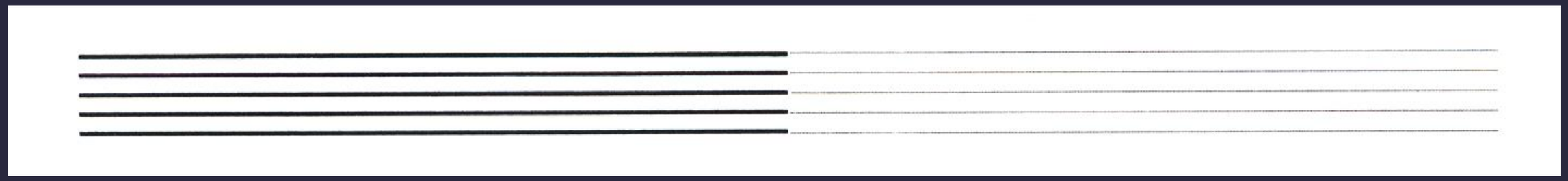


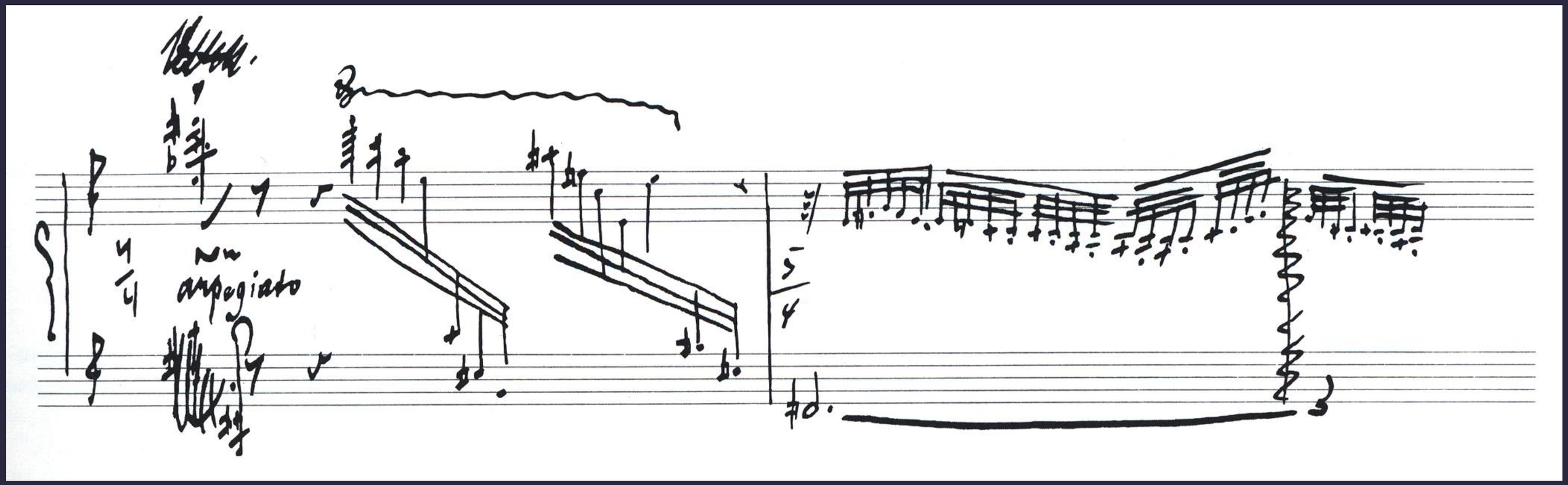
Signal and background compete above, as an electrocardiogram tracelines becomes caught up in a thick grid. Below, the screened-down grid stays behind traces from each of 12 monitoring leads:[4]



Electrocardiogram tracelines [from Tufte 90]

Layering: Gridlines

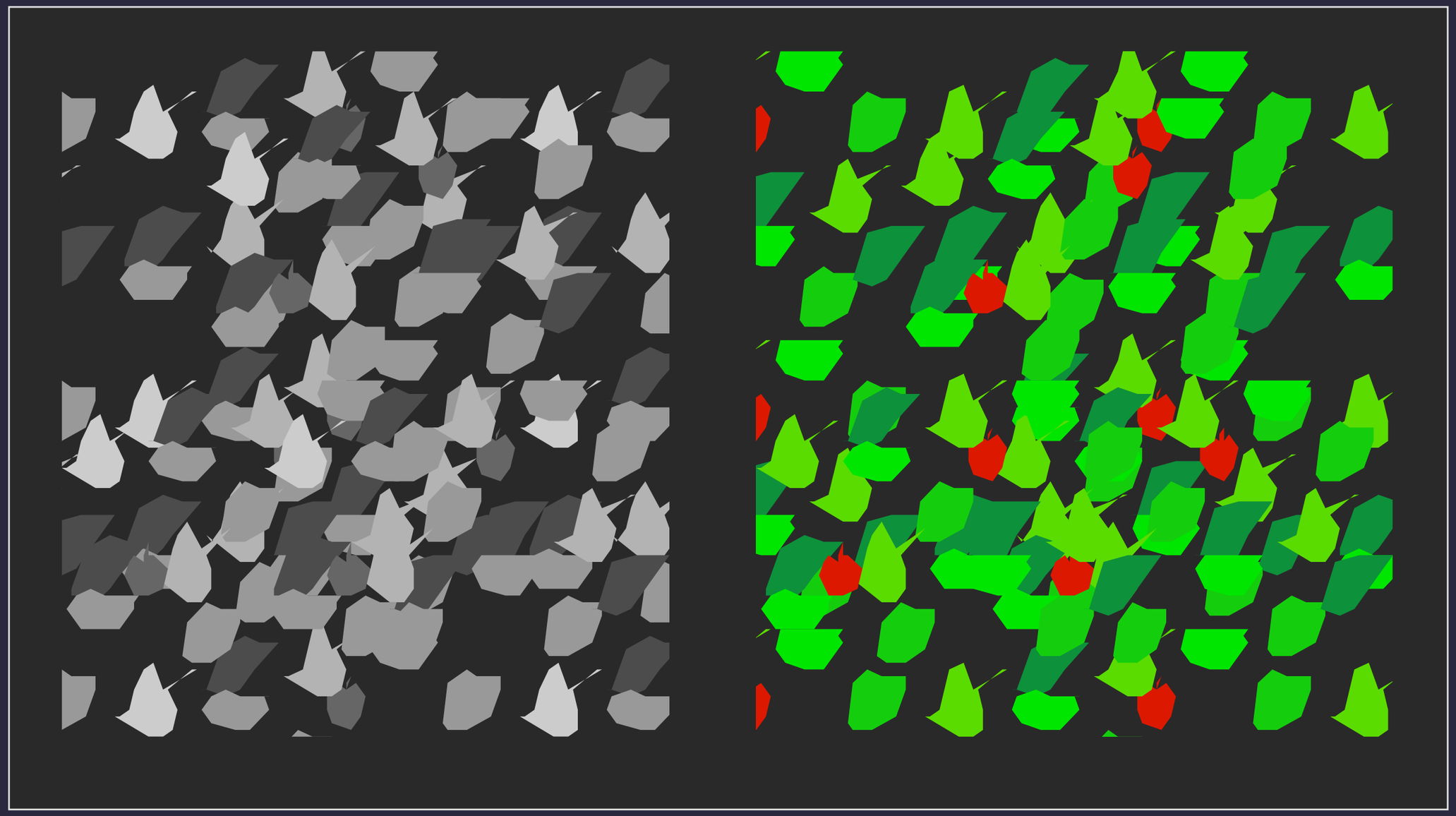




Stravinsky score [from Tufte 90]

Color in Visualization

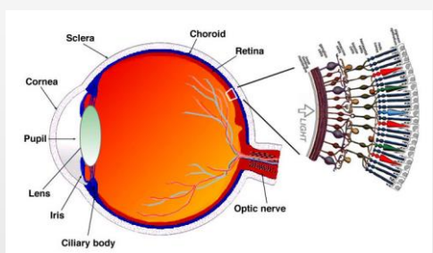
Identify, Group, Layer, Highlight



Colin Ware

What is Color?

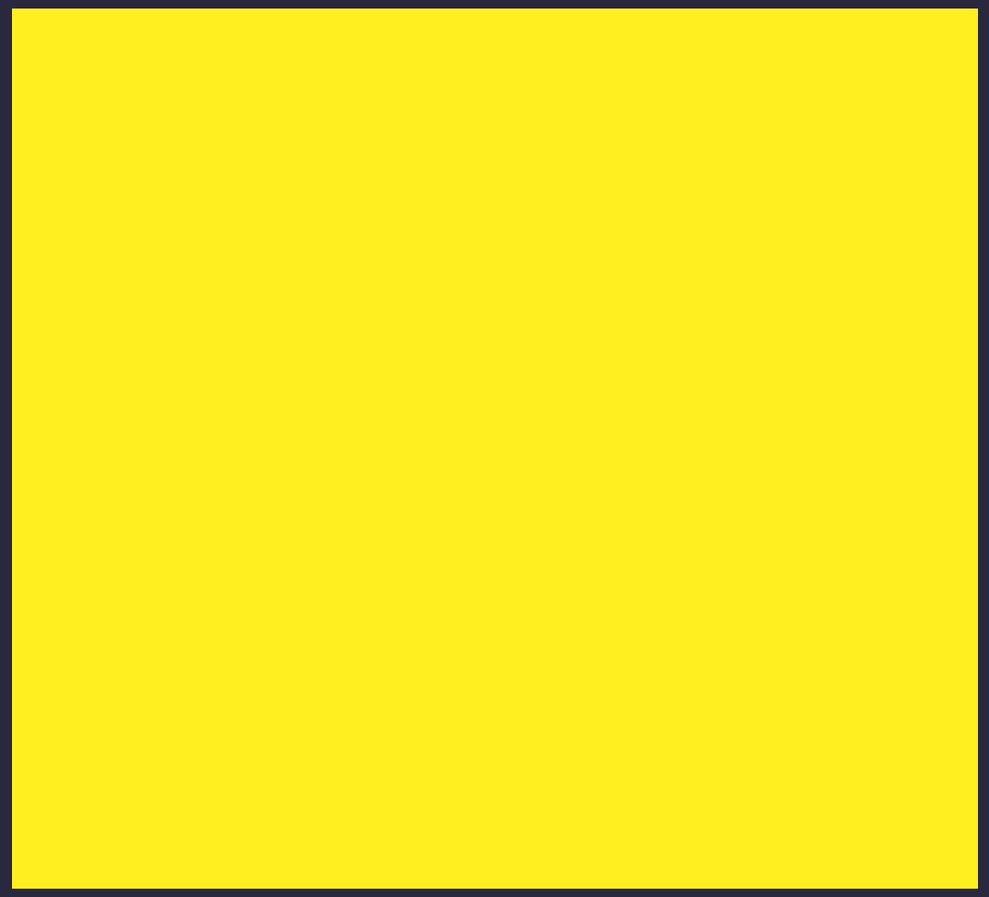
Physical World		Visual System		Mental Models
Lights, surfaces, objects	→	Eye, optic nerve, visual cortex	→	Red, green, brown Bright, light, dark, vivid, colorful, dull Warm, cool, bold, blah, attractive, ugly, pleasant, jarring



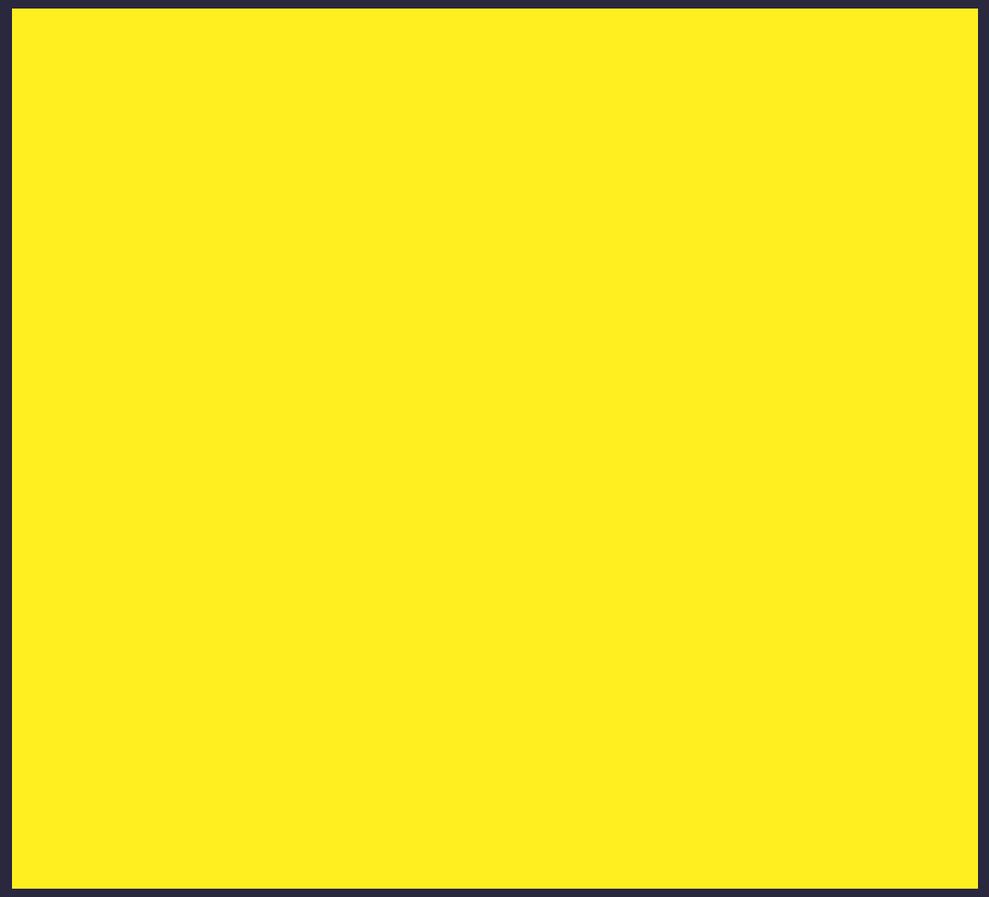
“Yellow”

Color Naming

What color is this?



What color is this?



“Yellow”

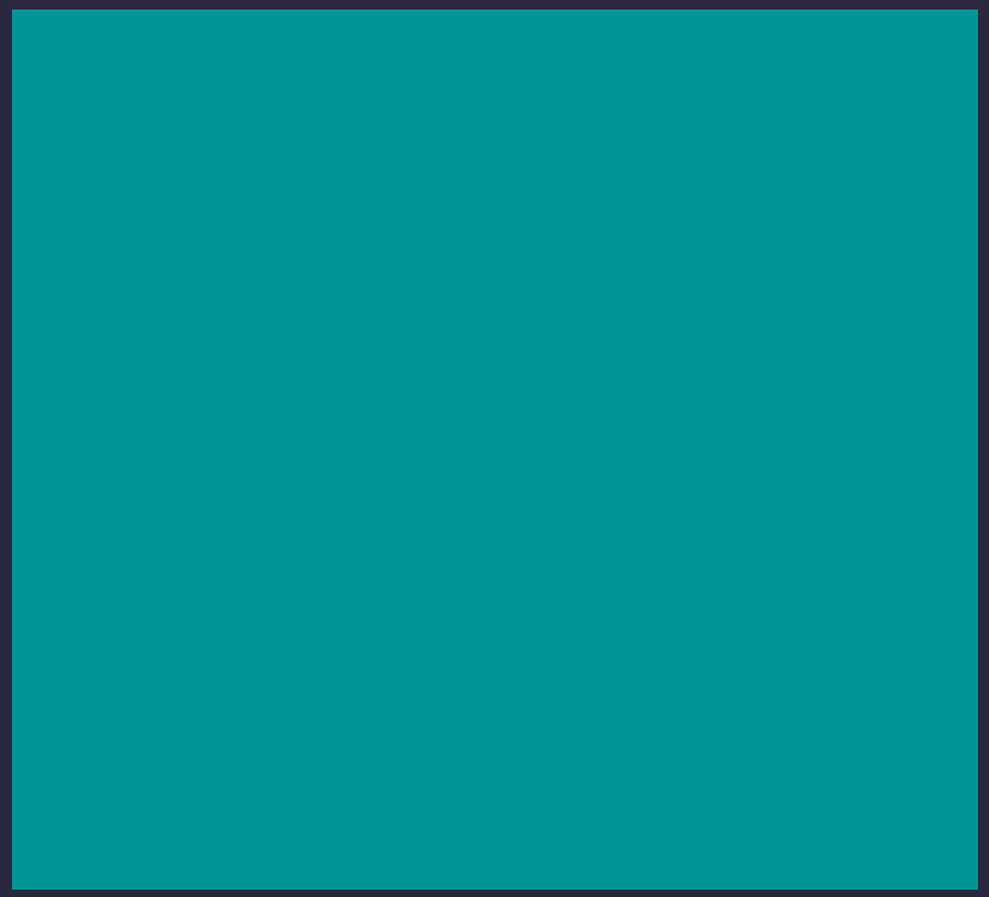
What color is this?

What color is this?

“Blue”

What color is this?

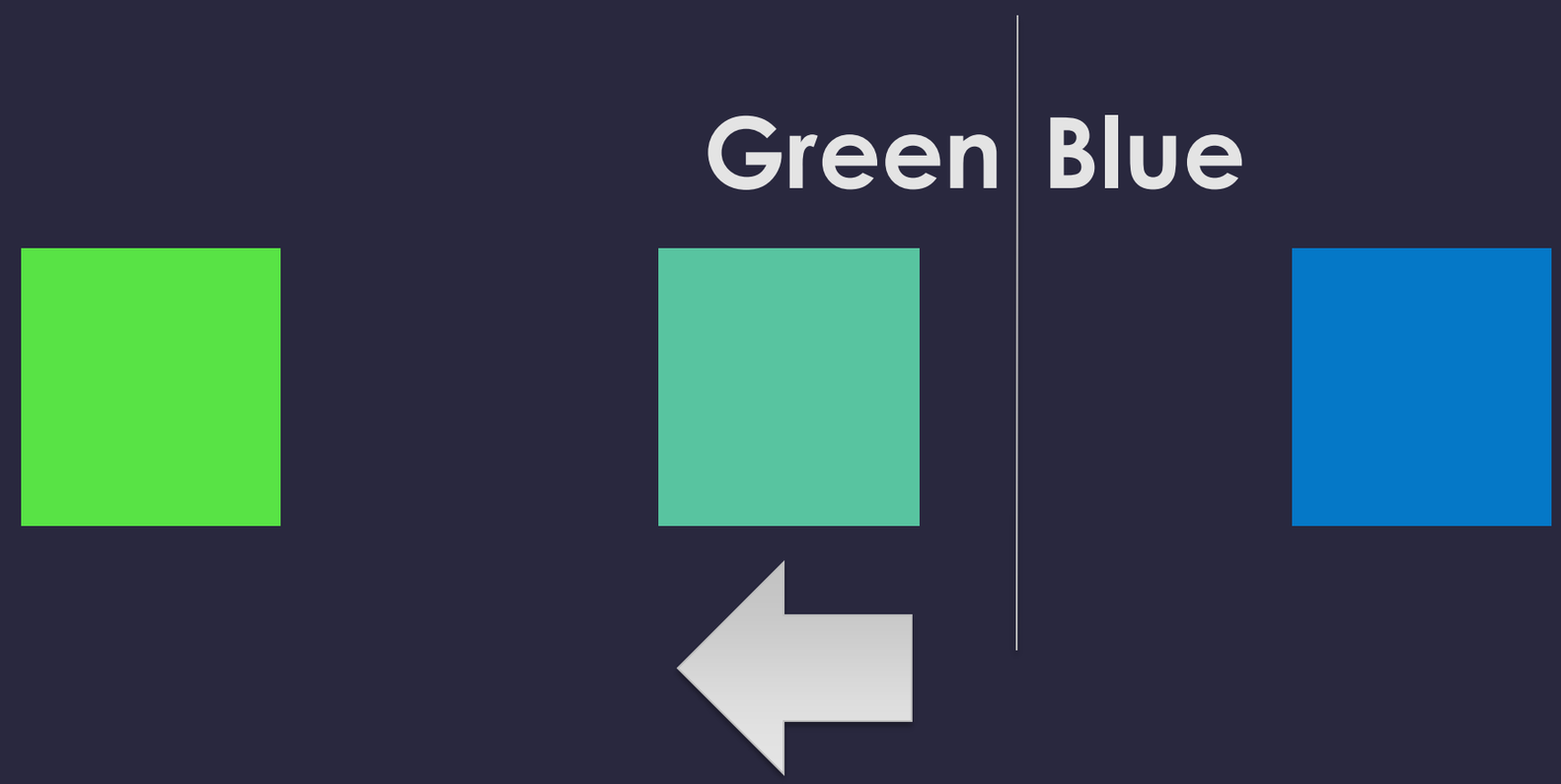
What color is this?



“Teal” ?

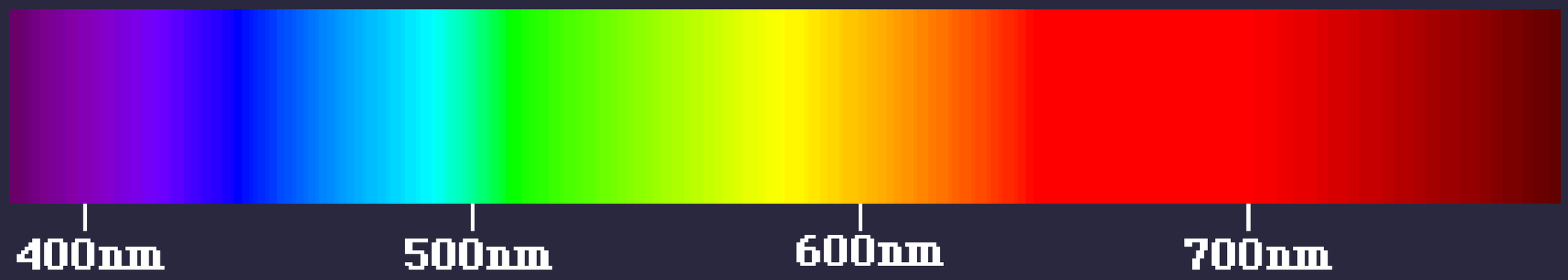
Naming affects color perception

Color name boundaries



Rainbow color ramp

We associate and group colors together, often using the name we assign to the colors



Rainbow color ramp

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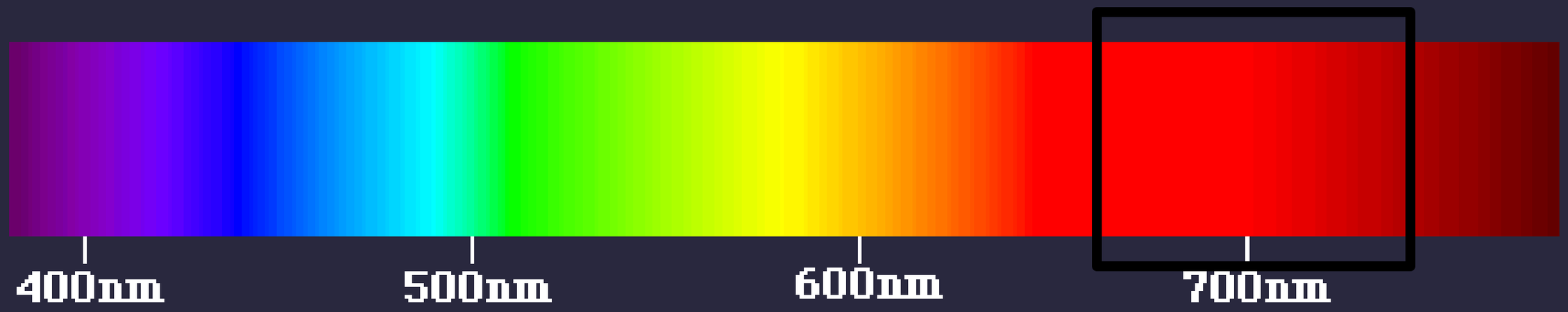
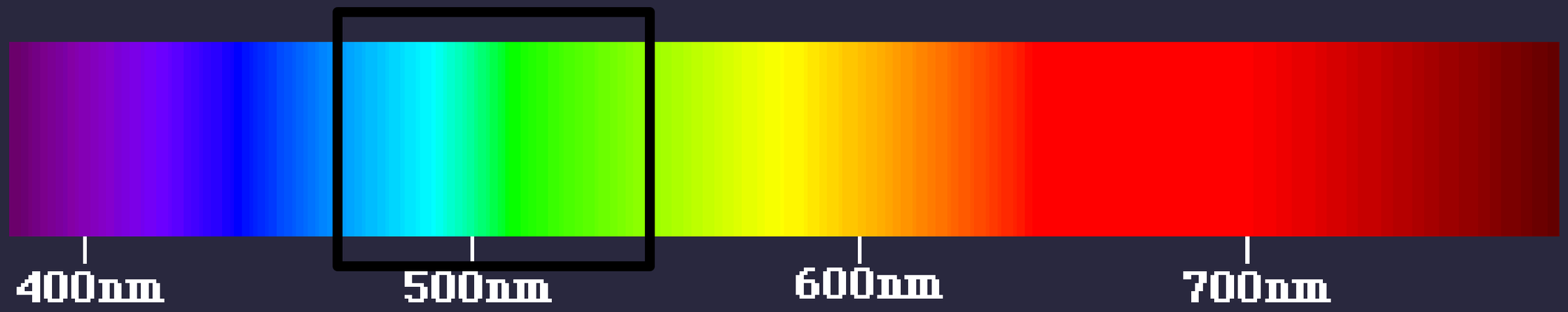


Rainbow color ramp

We associate and group colors together, often using the name we assign to the colors



Color naming models

[Heer & Stone]

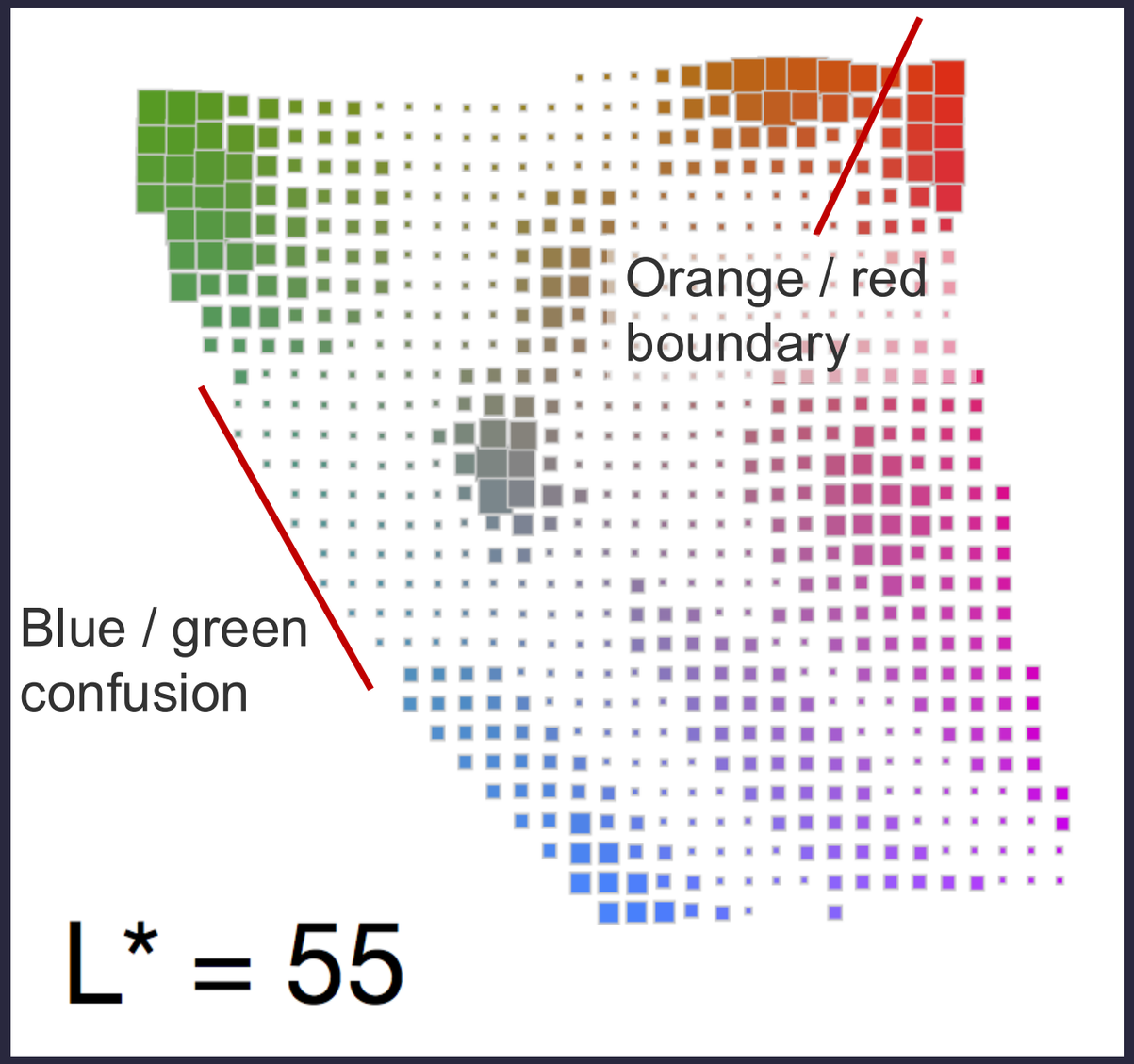
Model 3 million responses from XKCD survey

Bins in LAB space

sized by *saliency*:

How much do people agree on color name?

Modeled by entropy of *p(name | color)*



Palette Design + Color Names

Minimize overlap and ambiguity of color names

Color Name Distance										Salience	Name
0.00	1.00	1.00	1.00	0.98	1.00	1.00	1.00	1.00	0.20	.47	**blue** 62.9%
1.00	**0.00**	1.00	0.97	1.00	1.00	1.00	1.00	0.96	1.00	.90	**orange** 93.9%
1.00	1.00	**0.00**	1.00	1.00	1.00	1.00	1.00	0.90	0.99	.67	**green** 79.8%
1.00	0.97	1.00	**0.00**	1.00	0.95	0.99	1.00	1.00	1.00	.66	**red** 80.4%
0.98	1.00	1.00	1.00	**0.00**	0.96	0.91	0.97	1.00	0.99	.47	**purple** 51.4%
1.00	1.00	1.00	0.95	0.96	**0.00**	0.97	0.93	0.98	1.00	.37	**brown** 54.0%
1.00	1.00	1.00	0.99	0.91	0.97	**0.00**	1.00	1.00	1.00	.58	**pink** 71.7%
1.00	1.00	1.00	1.00	0.97	0.93	1.00	**0.00**	1.00	1.00	.67	**grey** 79.4%
1.00	0.96	0.90	1.00	1.00	0.98	1.00	1.00	**0.00**	1.00	.18	**yellow** 31.2%
0.20	1.00	0.99	1.00	0.99	1.00	1.00	1.00	1.00	**0.00**	.25	**blue** 25.4%
Tableau-10									*Average*	*0.97*	.52

http://vis.stanford.edu/color-

Palette Design + Color Names

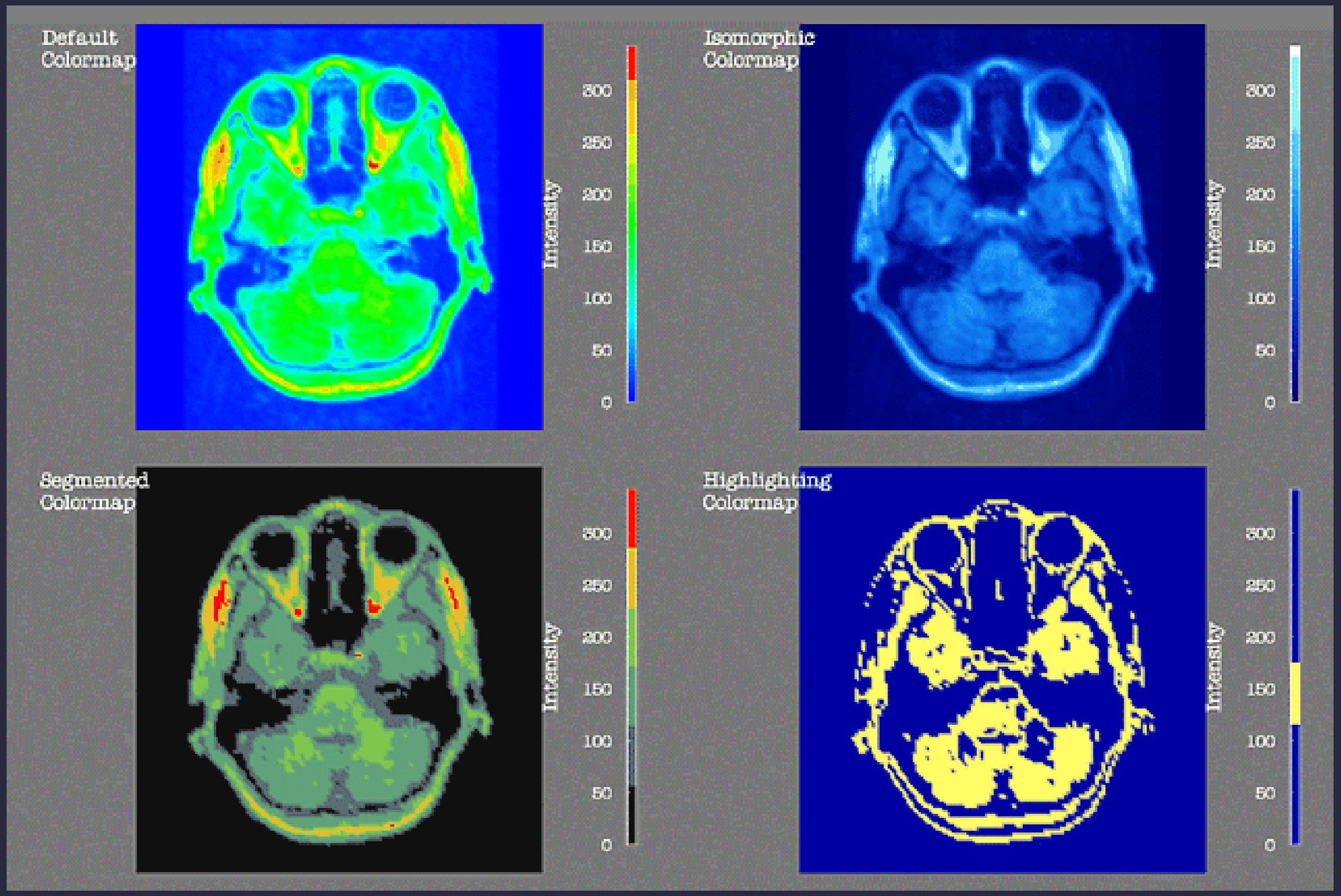
Minimize overlap and ambiguity of color names

Color Name Distance										Salience	Name
0.00	1.00	1.00	0.89	0.07	1.00	0.35	0.99	1.00	0.89	.30	**blue** 50.5%
1.00	0.00	0.99	1.00	1.00	0.92	1.00	0.84	0.98	0.99	.21	**red** 27.8%
1.00	0.99	0.00	1.00	0.98	1.00	1.00	1.00	0.17	1.00	.34	**green** 36.8%
0.89	1.00	1.00	0.00	0.98	1.00	0.71	0.93	1.00	0.32	.55	**purple** 67.3%
0.07	1.00	0.98	0.98	0.00	1.00	0.36	1.00	0.97	0.95	.20	**blue** 36.6%
1.00	0.92	1.00	1.00	1.00	0.00	1.00	0.97	0.99	1.00	.39	**orange** 51.9%
0.35	1.00	1.00	0.71	0.36	1.00	0.00	0.95	0.92	0.42	.13	**blue** 15.7%
0.99	0.84	1.00	0.93	1.00	0.97	0.95	0.00	0.98	0.85	.16	**pink** 29.4%
1.00	0.98	0.17	1.00	0.97	0.99	0.92	0.98	0.00	0.97	.12	**green** 21.7%
0.89	0.99	1.00	0.32	0.95	1.00	0.42	0.85	0.97	0.00	.30	**purple** 23.9%
Excel-10								*Average*	*0.87*	.27	

http://vis.stanford.edu/color-

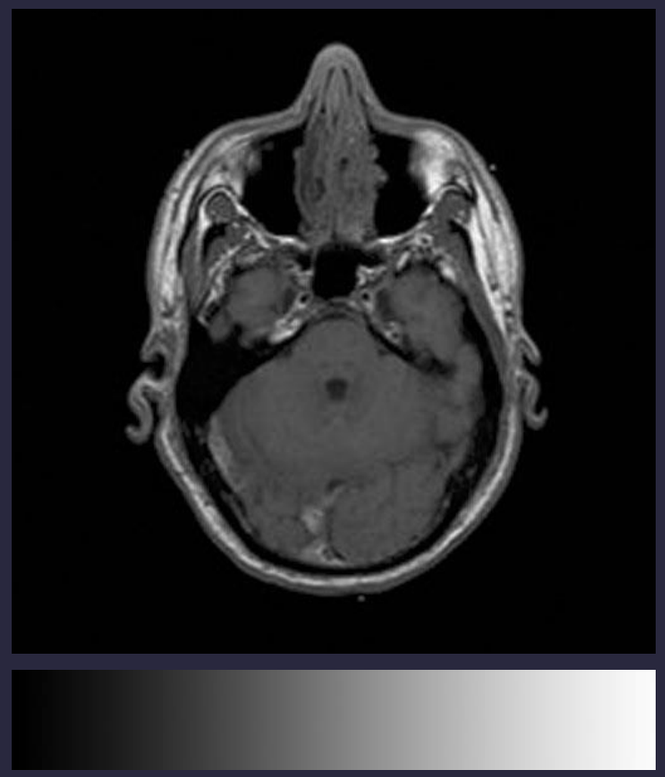
Mapping Data to Color Scale

Different Scales

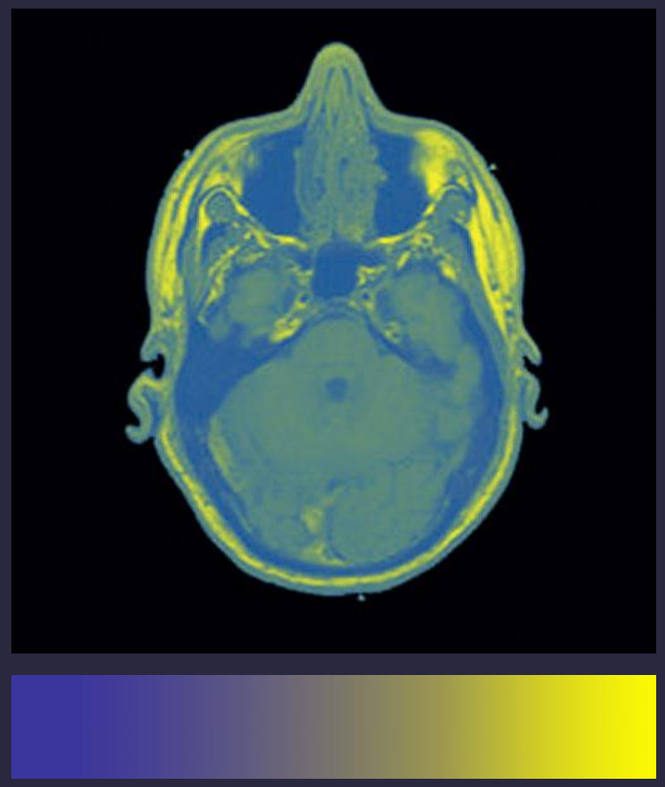


Rogowitz & Treinish, “How not to lie with visualization”

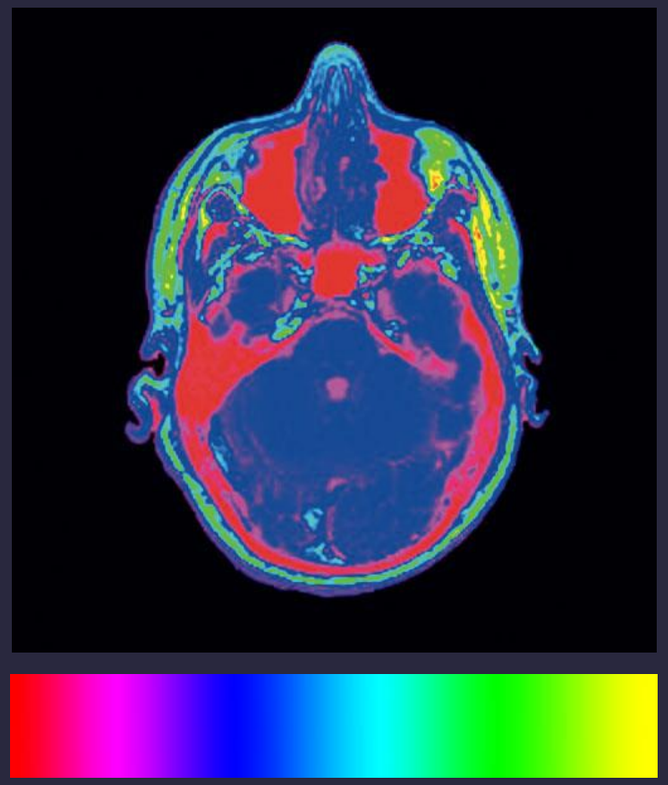
Different Scales



Lightness scale

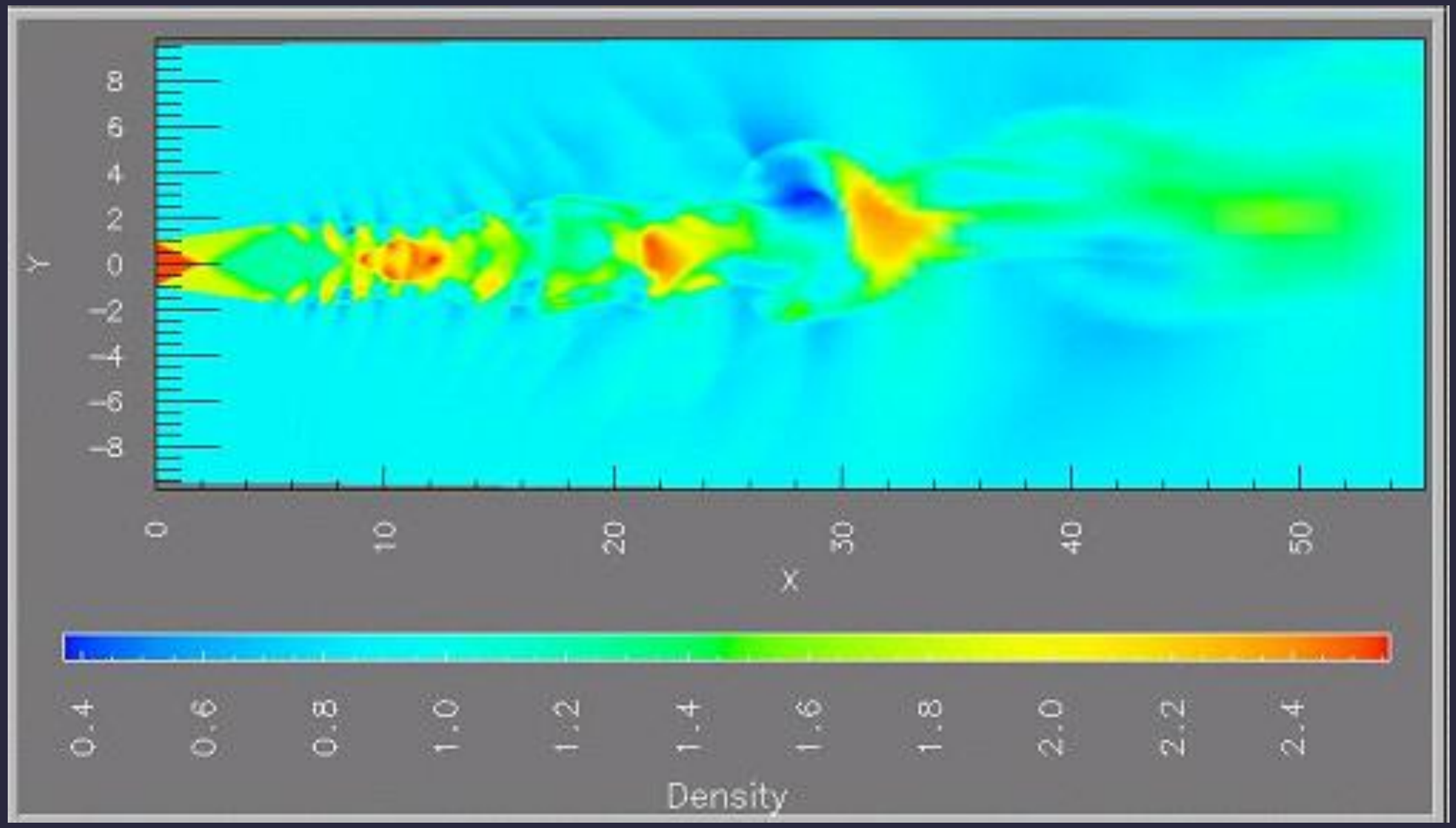


Lightness scale
with hue and
chroma variation



Hue scale with
lightness variation

Default rainbow maps



Phases of the Tides

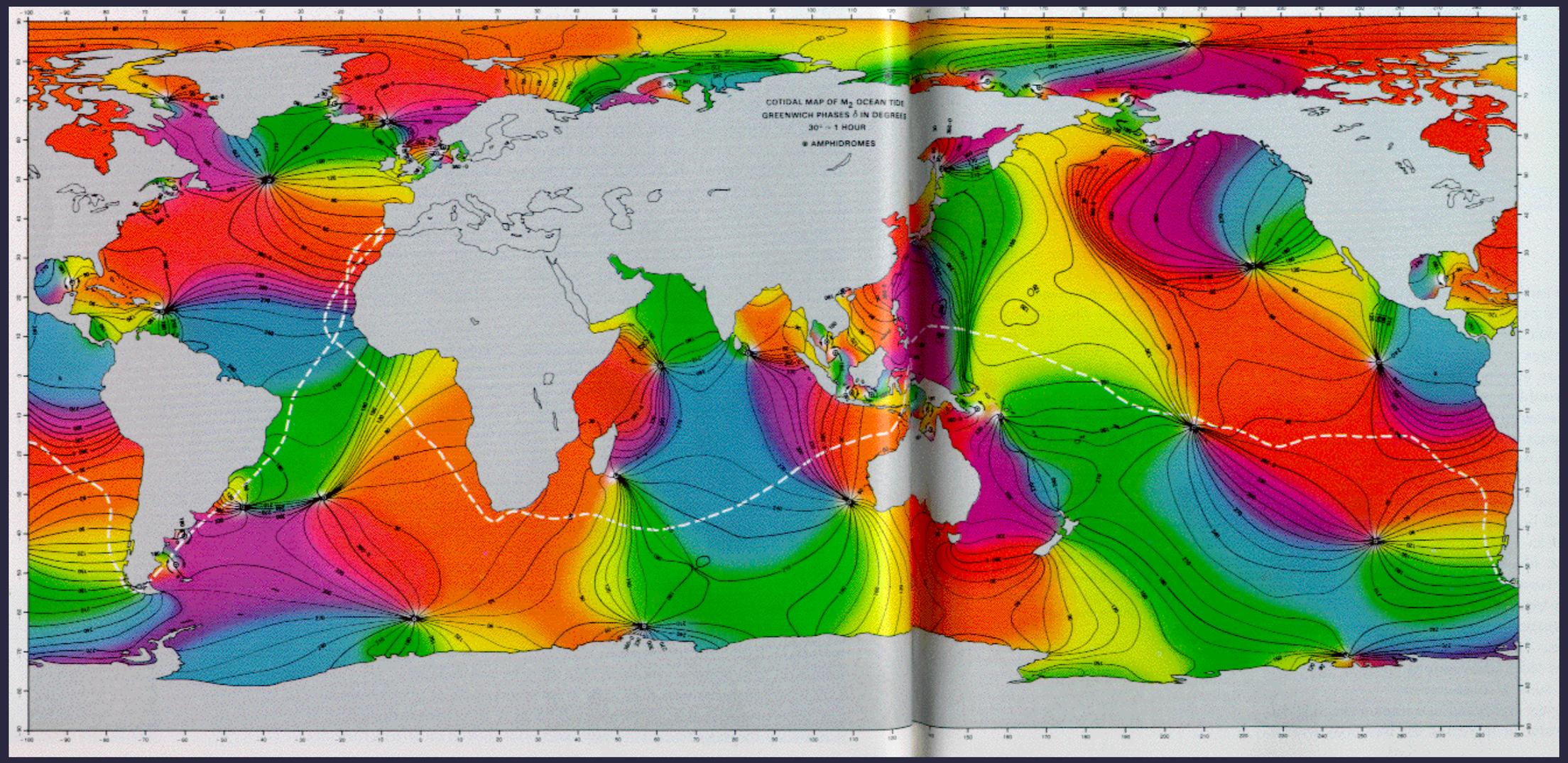


Figure 1.9. Cotidal chart. Tide phases relative to Greenwich are plotted for all the worlds oceans. Phase progresses from red to orange to yellow to green to blue to purple. The lines converge on anphidromic points, singularities on the earth's surface where there is no defined tide. [Winfree, 1987 #1195 , p. 17].

Quantitative color encoding

Sequential color scale

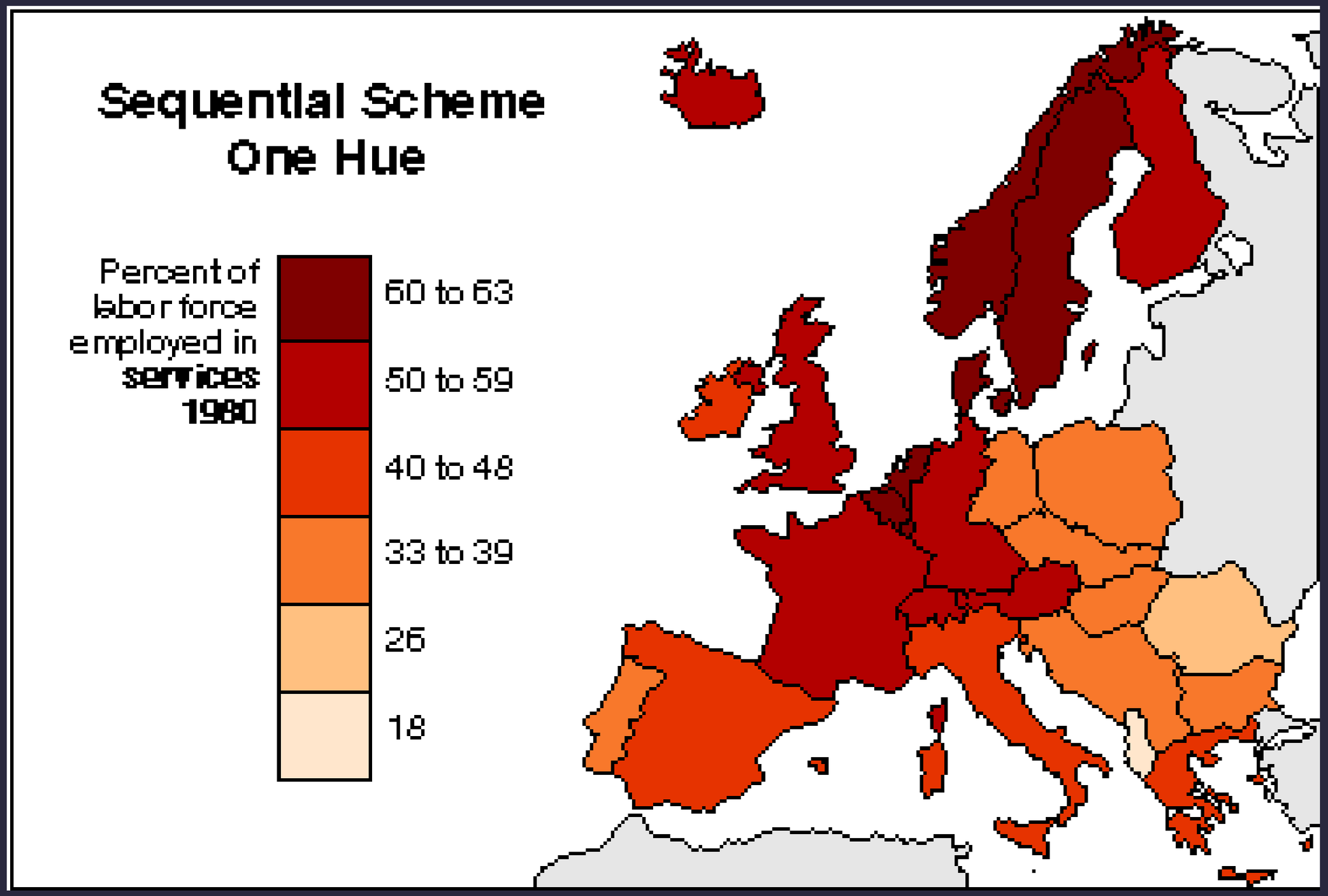
- Constrain hue, vary luminance/saturation
- Map higher values to darker colors

Diverging color scale

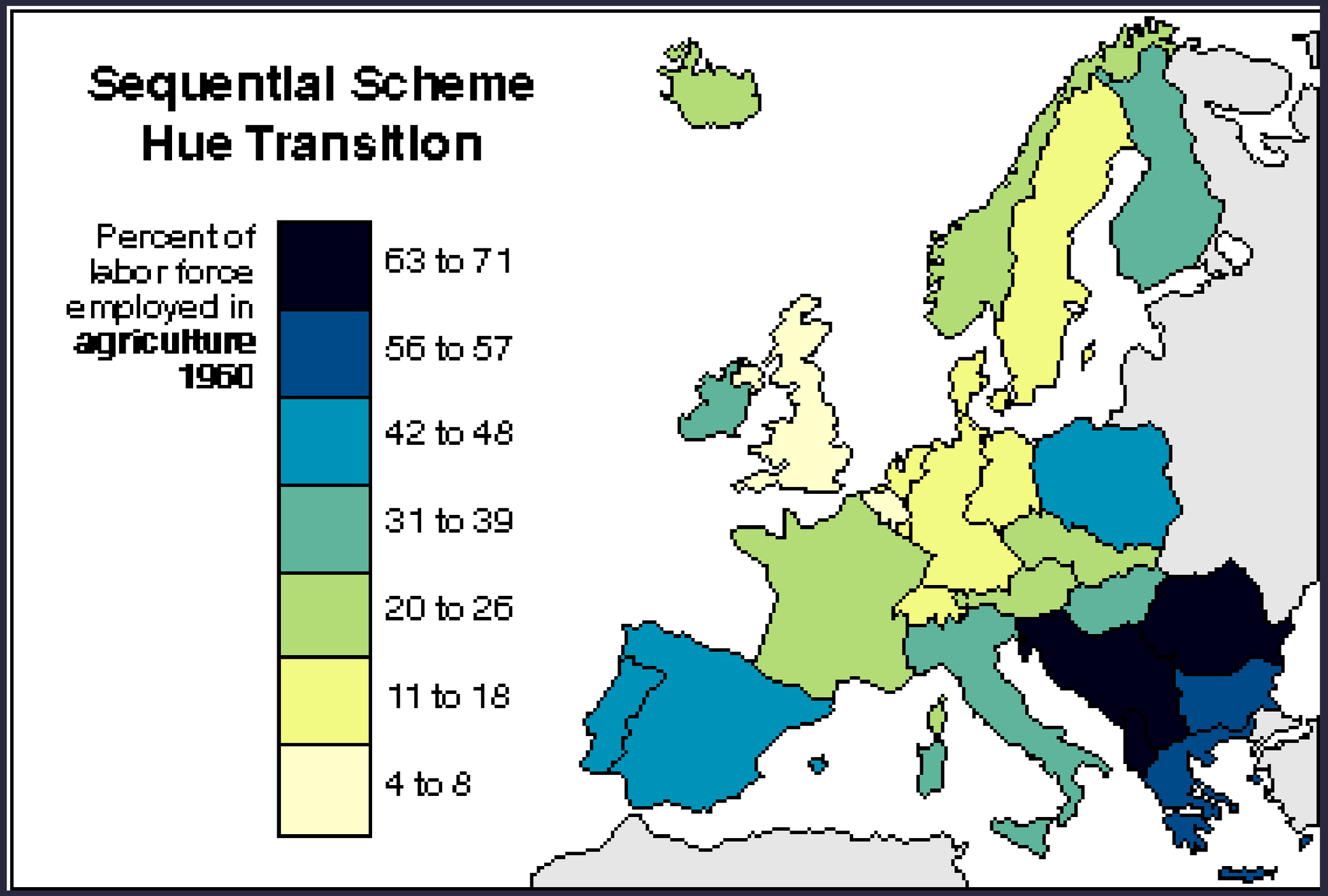
- Useful when data has a meaningful “midpoint”
- Use neutral color (e.g., grey) for midpoint
- Use saturated colors for endpoints

Limit number of steps in color to 3-9

Sequential color scheme



Sequential color scheme



Design of sequential color scales

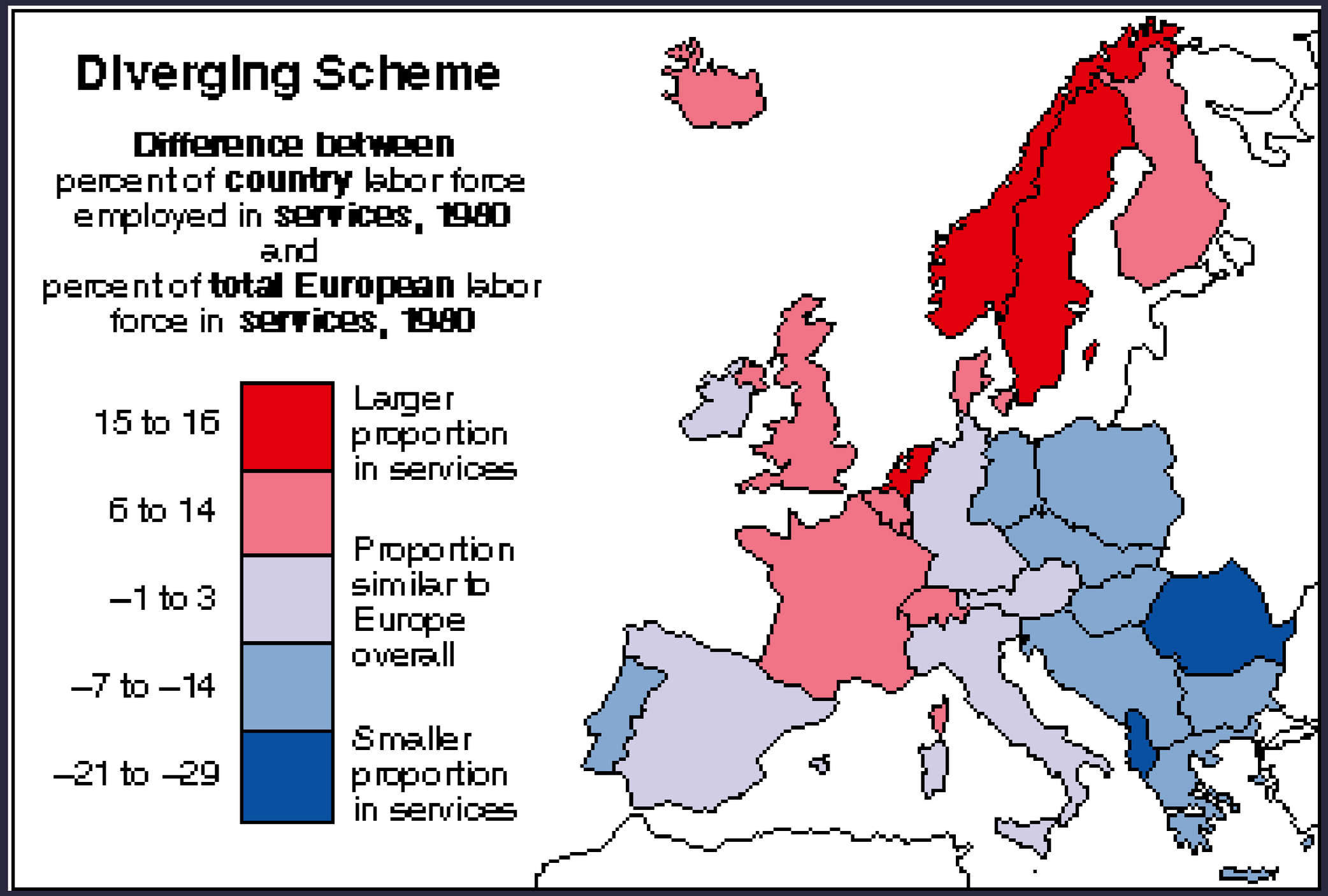
One Hue-Lightness *(Recommended)*

- Higher values mapped to darker colors
- ColorBrewer schemes have 3-9 steps

Hue Transition

- Two hues
- Neighboring hues interpolate better
- Couple with change in lightness

Diverging color scheme



Diverging color scheme

Hue Transition

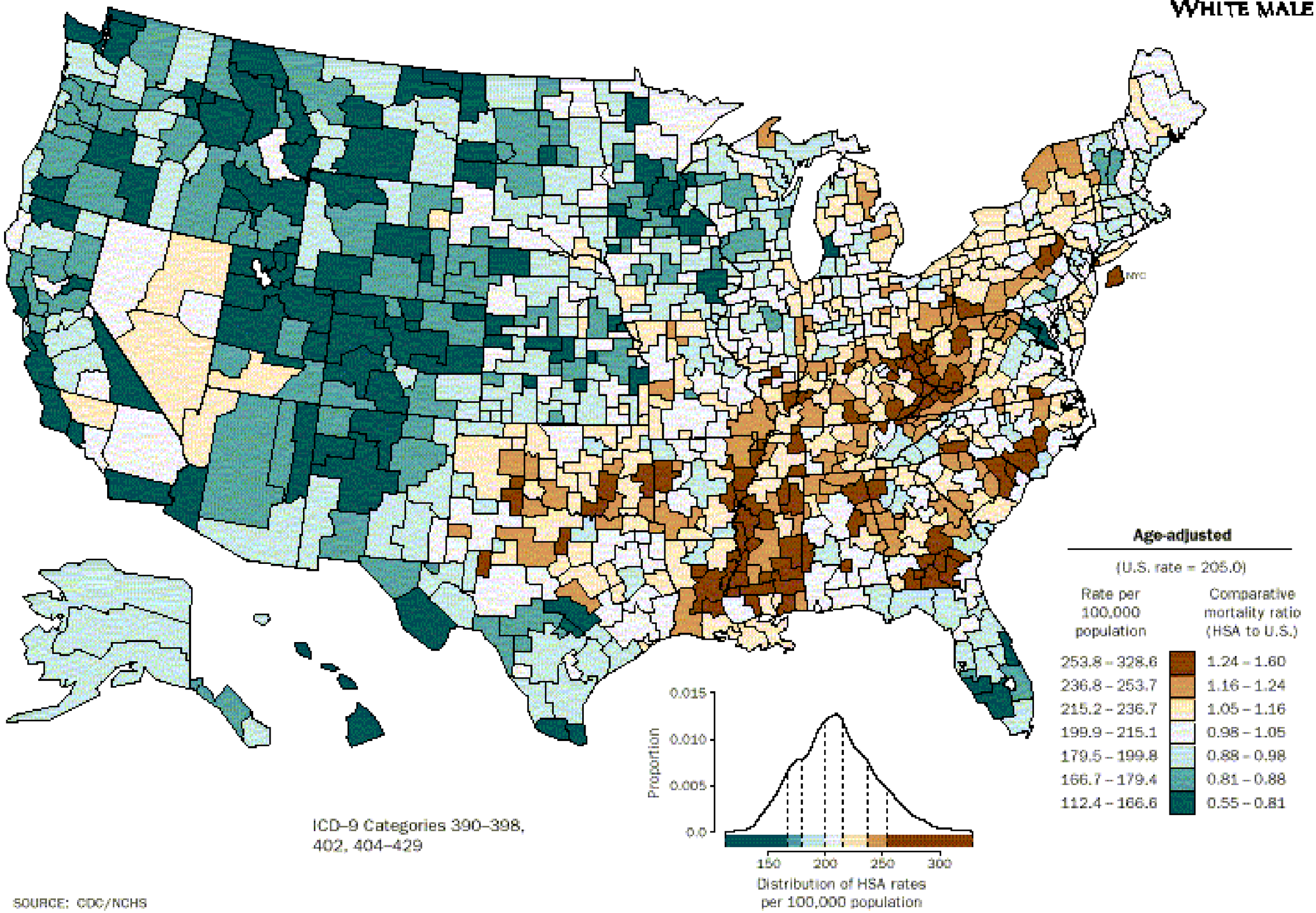
Carefully handle midpoint

- Quantizing data relative to average
  - Low, Average, High
  - ‘Average’ should be gray
- Otherwise, meaningful breakpoint
  - Defining middle value, e.g., 0 revenue
  - Positive & negative should use different hues

Extremes saturated, middle desaturated

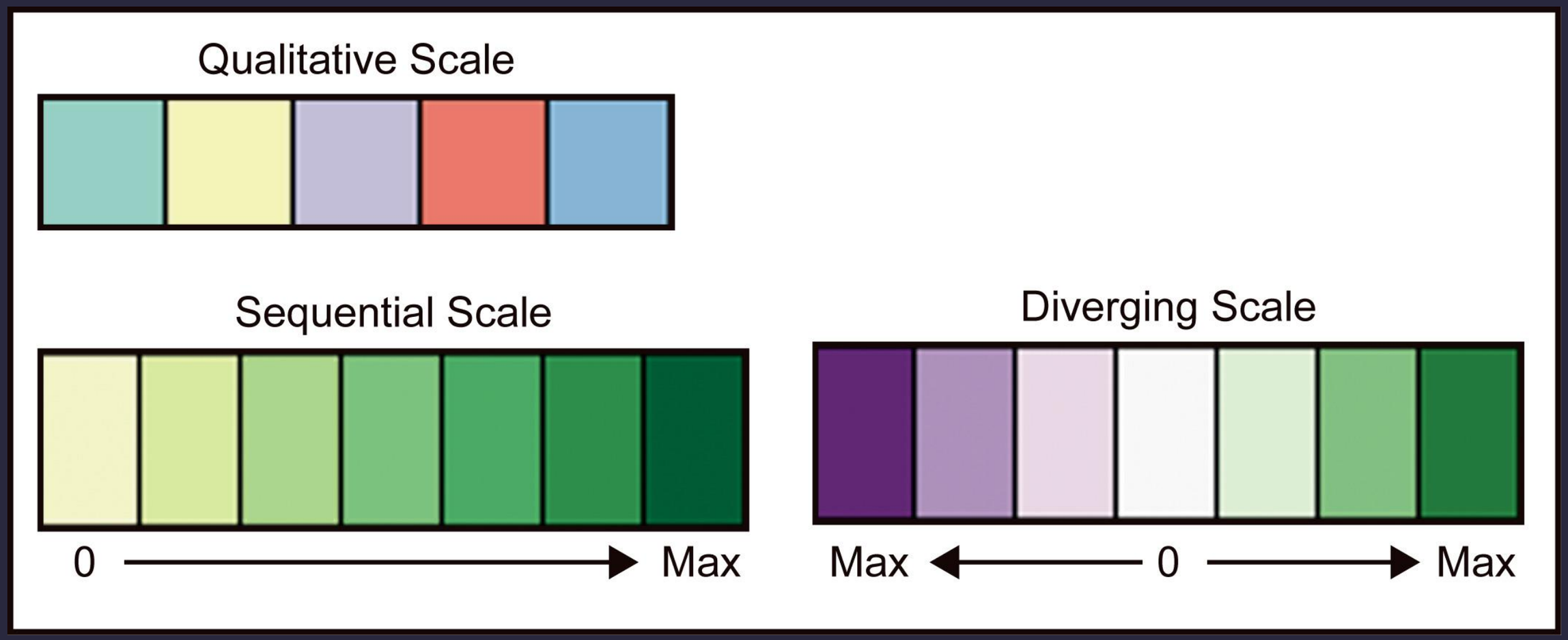
Age-adjusted death rates by HSA, 1988-92

Heart disease
White male



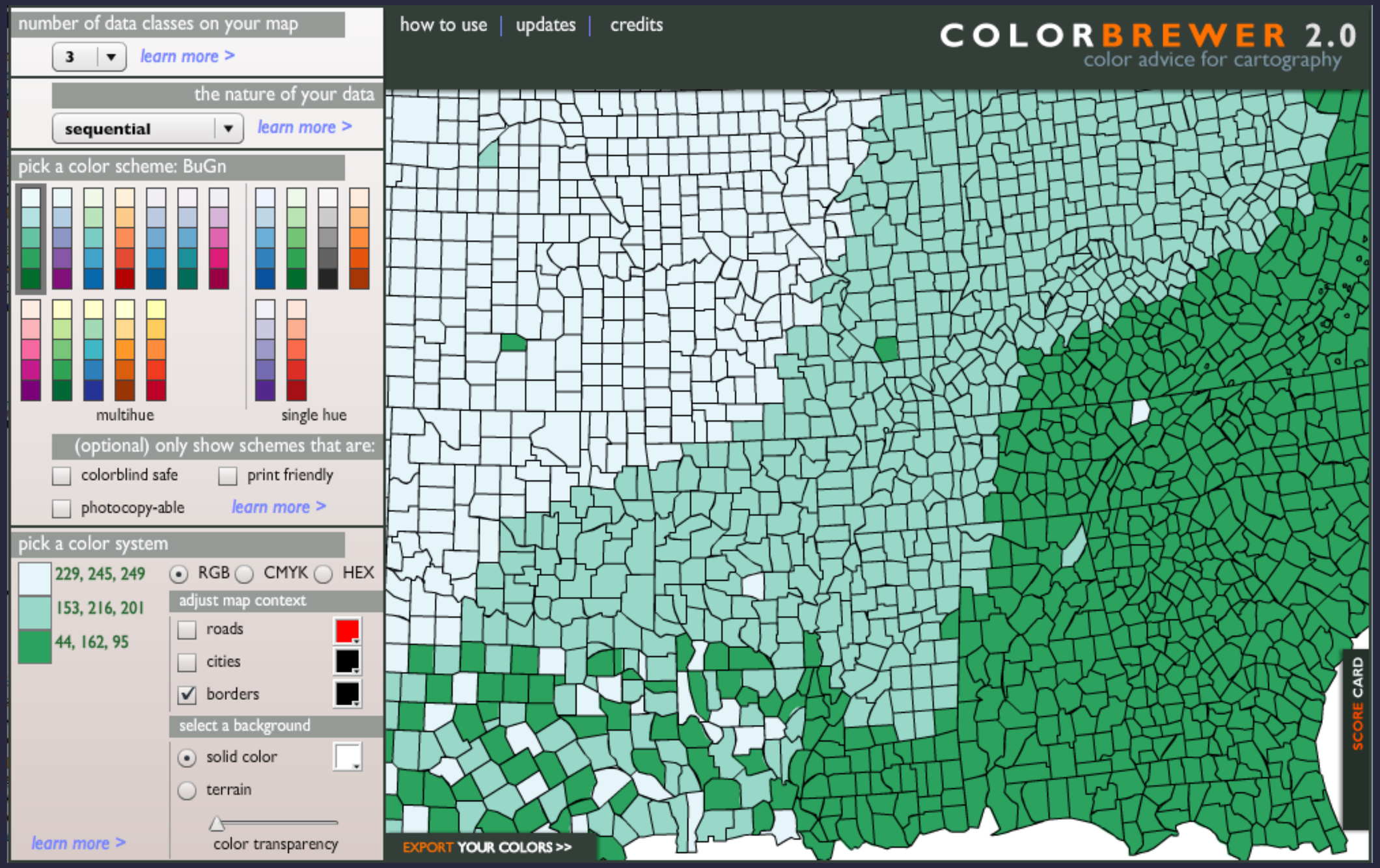
SOURCE: CDC/NCHS

Brewer's Categories



Cynthia Brewer, Pennsylvania State University

Color Brewer



www.colorbrewer.org

Summary: Color Design

Q/O variables: Control value (lightness)

- Ensure legibility
- Avoid unwanted emphasis
- Diverging: Design midpoint carefully

N variables: Use a limited hue palette

- Use distinctive, named hues
- Avoid too many competing colors
- Be aware of color blindness

Use neutral backgrounds

- Control impact of color
- Minimize simultaneous contrast